AI for Anomaly Detection

Lecture 2: Supervised Anomaly Detection





Lilach Goren Huber
Institute for Data Science (IDS)

zh aw School of Engineering

#	Date	Topic
1	15.09	Introducation to anomaly detection (AD)
2	22.09	Supervised ML for AD
3	29.09	Unsupervied ML for AD I
4	06.10	Unsupervised ML for AD II
5	13.10	Residual models: reconstruction ML methods
6	20.10	Residual models: reconstruction neural networks
7	27.10	Residual models: regression models
8	03.11	Initial assignment presentations – obligatory presence in person
9	10.11	Residual models: advanced regression neural networks (CNNs, TCNs)
10	17.11	Domain shifts and data scarcity: transfer learning
11	24.11	Contamination and mislabeling
12	01.12	Quantifying uncertainty, physics informed ML
13	08.12	Recent advances in AD research
14	15.12	Final assignment presentations – obligatory presence in person

Today

1. Introducing the course assignments

2. Recap last time

3. Supervised AD: logistics regression
   - Theory
   - Example

4. Supervised AD: Support Vector Machines (SVM)
   - Theory
   - Exercise
   - Solution

5. Your time to inspect the assignments and start grouping

Course grading

During the course you will progressively work on one assignment in groups of 3 students.

The assignment: anomaly detection on a selected dataset.

I. 20% → Initial assignment presentation (3.11).

II. 20% → Final assignment presentation + handing in of presentation and code (15.12).

In each presentation:

- 15-20min for each group.
- Each student is expected to participate **actively,** and answer questions.
- The grade is **individual per student.**

III. 60% → Exam on assignment presentation (January 2027).

- 10-15min for each student
- Each student is expected to know **all aspects of the assignment.**
- There will be questions on the **course content**, beyond your own assignment.
- By construction, each student will face **different questions** ("fairness" cannot be measured).

Course assignment

1. Split to teams of 3
2. Select a data set
   - You will have a selection of 8 data sets for anomaly detection
   - Different application fields (medicine, finance, retail, manufacturing, technical systems and machines).
   - Various data modalities (tabular, images, time-series).
3. Week 8 (3.11): presenting your data exploration and task definition (20%)
   - Download the data set
   - Prepare your explorative analysis (plot the raw data, correlations, first features, etc)
   - Define the anomaly detection task
   - Prepare a short presentation and the code for discussion at class
4. Week 14 (15.12): presenting your anomaly detection results (20%)
   - Prepare a short presentation (15 min)
   - Explain your approach to anomaly detection and results
   - Demonstrate how you applied the methods learned during the lectures.
   - Hand in your slides and code.
5. Exam (individual): your project and beyond (60%)
   - Questions about all aspects of your project
   - Questions about the material learned at class

During the lectures there will be **dedicated time slots** in which you will be able to get feedback to your assignments

Course assignment

Work in groups of 3 students

- Choose one of the provided datasets and investigate it as an anomaly detection problem.
- Your objective is **not** to obtain the highest score, but to design and conduct a convincing anomaly detection study.

Requirements

- Define your own anomaly detection question(s).
- Motivate the chosen methods and compare **at least 3 methods learned in class**.
- Include **at least one additional method** not covered in the course.
- If your dataset was used during the lectures (e.g. FMNIST or Turbofan), your project must go **substantially beyond** the classroom examples.
- Support your conclusions with appropriate visualizations, evaluation metrics, and systematic experiments.
- Critically discuss the strengths and limitations of your approach.
- Convince me that you (and not your favourite LLM) understood anomaly detection.

Course assignment

You decide…

- Which methods are worth comparing
- Which preprocessing is appropriate
- Which evaluation protocol is meaningful
- Which hyperparameters deserve tuning
- Which model improvements are worth investigating
- What conclusions can (and cannot) be drawn from the results
- There is **no single correct solution.** A well-designed investigation is valued more than a complex model.

Evaluation criteria

1. Problem formulation and motivation
2. Experimental design and methodology
3. Quality and fairness of model comparison
4. Analysis, visualization and interpretation
5. Critical discussion and conclusions
6. Novelty beyond the course examples

→ The goal is NOT to reach SOTA results but to show you understood the concepts learned in the course.

Data set 1: Detecting malignant breast cancer tumors (WDBC)

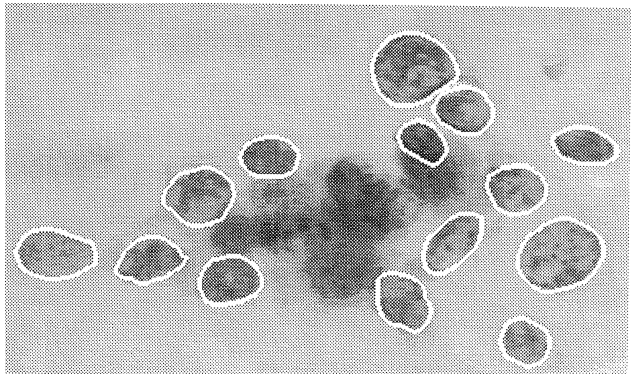
Traditional surgical tissue biopsies are highly invasive, costly, and stressful for patients. The Wisconsin Diagnostic Breast Cancer (WDBC) task aims to bypass this by training models to analyze cell patterns from a minimally invasive Fine Needle Aspirate (FNA) fluid sample, enabling rapid, automated, and highly accurate differentiation between benign and malignant tumors.

Data: 569 patient samples:

Input Features: 30 real-valued numerical attributes derived from digital scans of cell nuclei.

Feature Construction: 10 distinct morphological traits of the cell nuclei are measured. Aggregations: For each trait, the program calculates the Mean, Standard Error, and Worst (Largest) value across the image → 30 features.

Target Variable: Malignant (M) (212 cases) vs. Benign (B) (357 cases)



cell contours used for feature computation

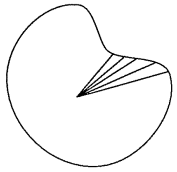


Figure 3: Radial Lines Used for Smoothness Computation



Figure 4: Chords Used to Compute Concavity

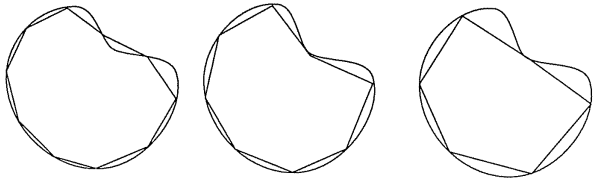


Figure 6: Sequence of Measurements for Computing Fractal Dimension

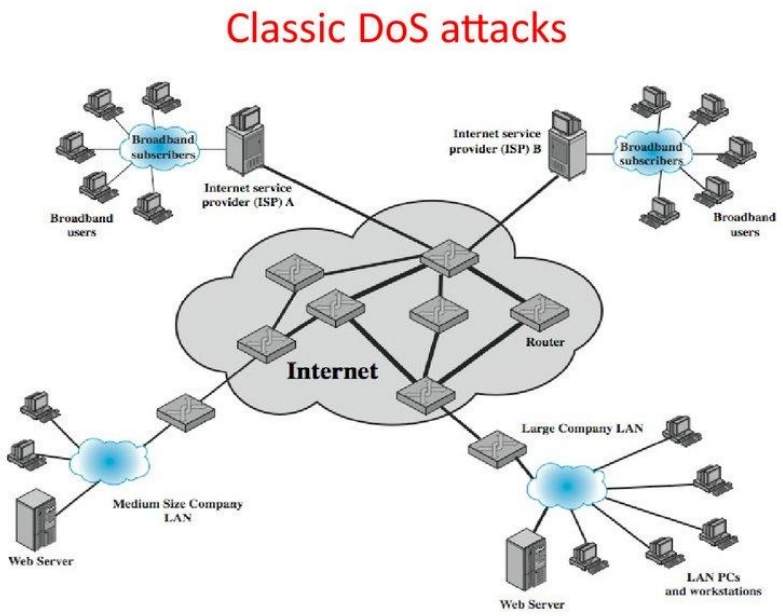
"Nuclear feature extraction for breast tumor diagnosis", Nick Street at al. 1993.

dataset https://archive.ics.uci.edu/dataset/17/breast%2Bcancer%2Bwisconsin%2Bdiagnostic?

Data set 2: Network intrusion detection (UNSW-NB15)

Network intrusion detection aims to identify malicious network activity by distinguishing it from normal communication patterns.

- Detect Denial-of-Service (DoS) attacks from network traffic.
- Normal traffic: legitimate communication between hosts
- DoS attack: an attacker floods a server with requests, preventing legitimate users from accessing the service
- We treat normal traffic as the normal operating condition and DoS traffic as anomalies.



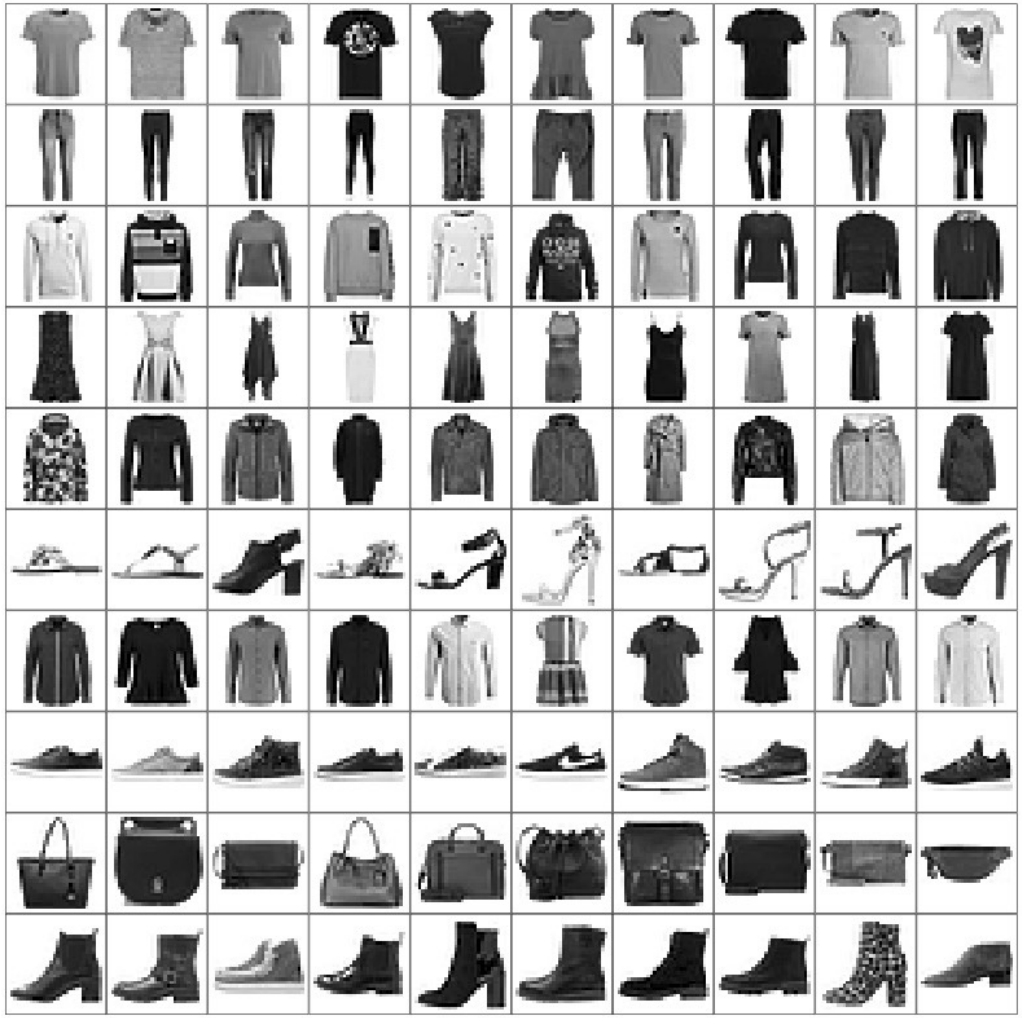
Connection	dur	dpkts	dbytes	rate	state	Class
1	0.12	6	932	117	CON	Normal
2	0.04	3	128	10325	INT	DoS
3	0.18	9	1410	111	CON	Normal
4	0.03	2	96	13067	INT	DoS
5	0.27	13	1854	104	CON	Normal

Feature	Description
`dur`	Connection duration
`dpkts`	Destination → source packet count
`dbytes`	Bytes sent by destination
`rate`	Packet rate
`state`	Connection state

Data set 3: Detecting wrong fashion article images (F-MNIST)

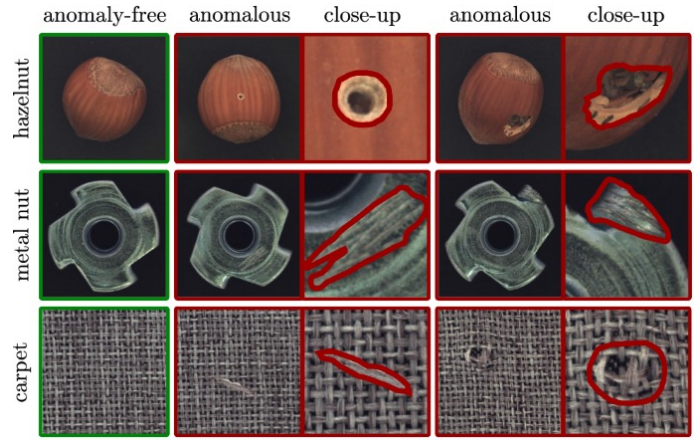
- Created by Zalando
- 10 fashion article classes
- 60000 training images
- 10000 test images
- Typically used for classification benchmarking.
- Can be used for image anomaly detection benchmarking.
  - E.g trousers as normal class, others as abnormal.
- In lecture 5 we will work with it.

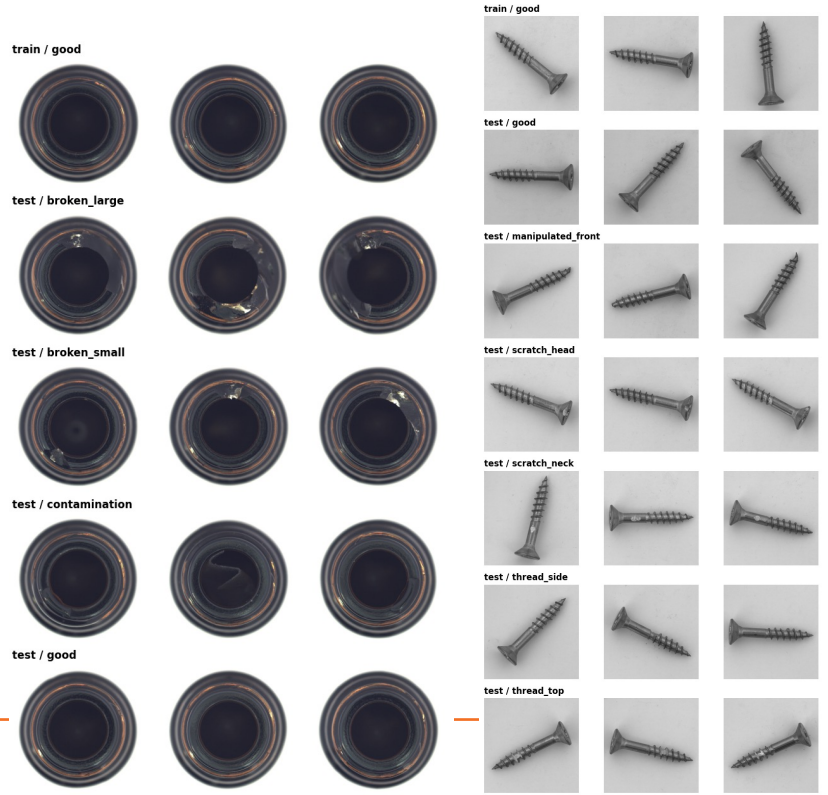
→ Your task design will have to go considerably beyond this.



Data set 4: Detecting defects in industrial images (MVTec AD)

- Industrial inspection: detect defects reliably before faulty products reach customers or damage downstream processes.
- Goal: Automate visual inspection for faster and scalable quality control than purely manual inspection.
- The **MVTec Anomaly Detection (MVTec AD)** created by MVTec Software GmbH as a benchmark for visual AD for industrial inspection.
- High-resolution color images from multiple object and texture categories.
  - training set defect-free images
  - test set defect-free images and images with different types of defects.
  - Annotations: per image and per pixel (for segmentation)
- Your task: focus on bottles and screws.





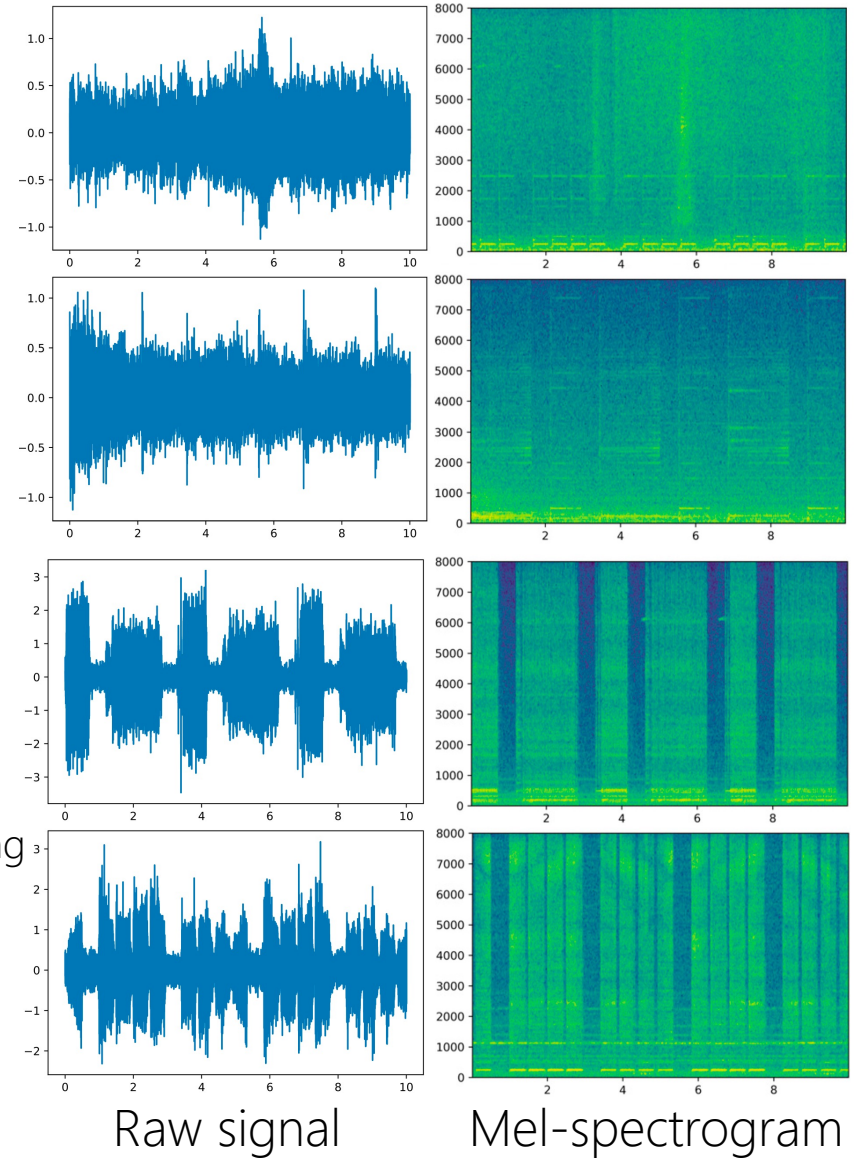
Data set 5: Detecting malfunctioning industrial machines using audio signals (MIMII)

- Audio recordings of industrial machines operating under **normal** and **faulty** conditions.
- Goal: detect malfunctioning machines.
- Here: only industrial fans
  - 4 different physical fan units.
  - A single microphone per fan.
  - 3 different background-noise conditions.
  - 10-second audio recordings sampled at 16 kHz.
  - normal and anomalous recordings,
- Design your own AD setting and experiment.

- Note: data is split into 3 zip files on Moodle.

Purohit (2019) "MIMII Dataset: Sound dataset for malfunctioning industrial machine investigation and inspection"





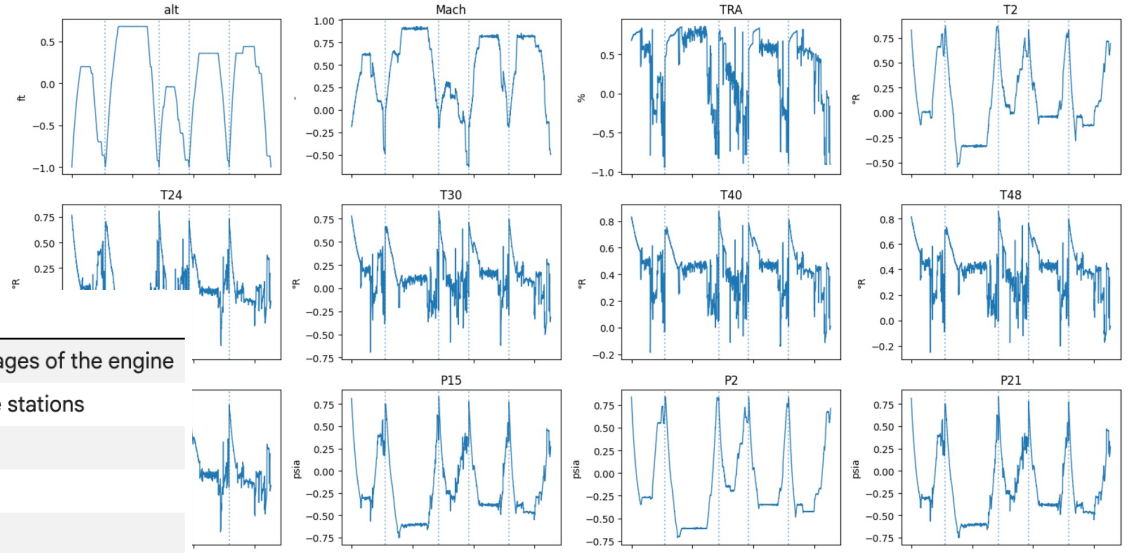
Raw signal Mel-spectrogram

Data set 6: Detecting degradation of aircraft engines (NASA CMAPSS)

- The NASA N-CMAPSS dataset contains **simulated turbofan-engine measurements** generated with a high-fidelity physics-based model.
- Each engine starts in a healthy condition and gradually degrades. In contrast to an abrupt fault, gradual degradation is better described by a **continuous anomaly score** than by only a binary normal/abnormal decision.



Variable	Description
alt	Flight altitude
Mach	Aircraft Mach number (speed relative to the speed of sound)
TRA	Throttle Resolver Angle (pilot throttle setting)
T2	Ambient inlet air temperature



Variable	Description
T24, T30, T40, T48, T50	Temperatures measured at different stages of the engine
P15, P2, P21, P24, Ps30, P30, P40, P50	Pressures measured at different engine stations
Nf	Fan rotational speed
Nc	Core compressor rotational speed
Wf	Fuel flow rate

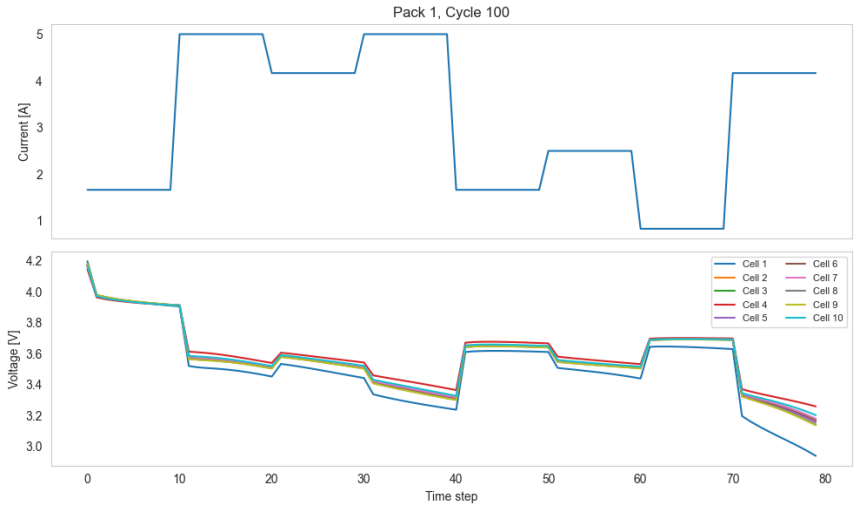
In lecture 7 we will use this dataset.

→ Your task design will have to go **considerably beyond** this.

Data set 7: Detecting accelerated battery aging

- Background
  - Li-ion batteries are organized into packs
  - Each pack consists of multiple interconnected cells
  - Cells naturally degrade over time and usage
  - Some cells may exhibit abnormal degradation
- Goal: early detection of accelerated degradation.
- Why?
  - Safety: Prevent fire hazards and thermal runaway
  - Costs: extend battery lifetime by replacing individual cells.
- Data:
  - Discharge cycles: current and voltage vs. time.
  - Full degradation trajectory: 100 cycles per cell.
  - 10 cells per pack, connected in series.



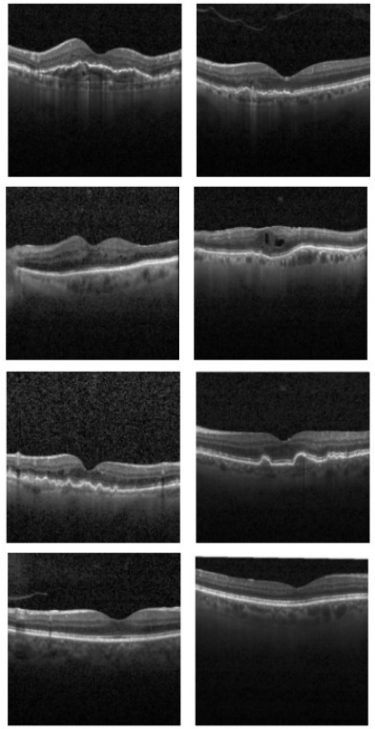


Data set 8: Detecting retinal diseases from images (OCTMNIST)

Optical coherence tomography (OCT) is a non-invasive imaging technique that uses light to obtain high-resolution cross-sectional images of tissue. OCT is routinely used to visualize the layered structure of the retina.

Manually reviewing thousands of optical OCT slices places an immense cognitive load on eye care specialists. Automated DL models trained on data like OCTMNIST act as a **clinical triage tool**. Rapid flagging of abnormal scans ensures patients with aggressive diseases receive urgent, vision-saving specialist interventions.

- **CNV**: Leaky abnormal blood vessels grow into the retina.
- **DME**: High blood sugar causes fluid leakage and swelling.
- **Drusen**: Fatty protein deposits accumulate beneath the retina.
- **Normal**: Healthy retinal tissue layers with no structural damage.



"Identifying Medical Diagnosis and Treatable Diseases by Image-Based Deep Learning", Kermany et al. 2018.

Course assignment: summary and timeline

	Dataset	Data modality	Seen at class?	Data loader
01	Breast tumors	Tabular (num)	No	Loader from sklearn.datasets
02	Network intrusion	Tabular (num+cat)	No	Loader from public link
03	Fashion articles	images	yes	Loader from public link
04	Industrial items	images	No	Local data + loader
05	Machine audio	Timeseries (univar.) or spectrograms	No*	Local data + loader
06	Aircraft engines	Timeseries (multivar.)	yes	Loader from public drive
07	Battery packs	Timeseries (bivar.)	No	Local data + loader
08	Retinal disease	images	No	Local data + loader



Next class:

your time to ask questions about the datasets.

3.11:

be ready with a first data exploration and task definition.

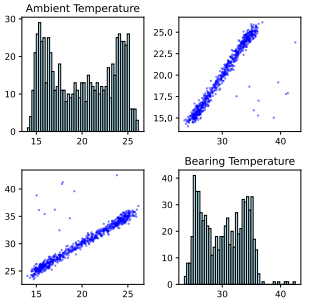
15.12:

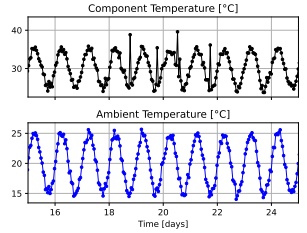
be ready with a final results presentation.

2026_student_assignment_list.xlsx

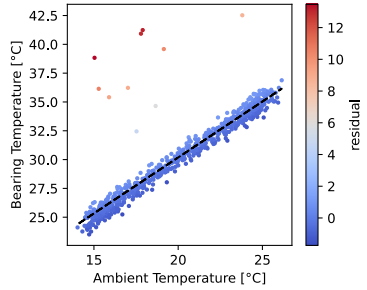
Recap: AD with statistical thresholds

- Step 1: **visualize** raw variables.
- Step 2: **visualize** correlations.
- Step 3: model the "usual" behaviour.
  - Linear regression
  - Robust linear regression → inherently better for AD

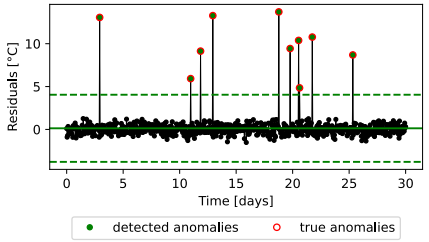




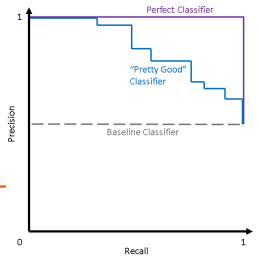
- Step 4: calculate residuals and use them to derive anomaly scores.

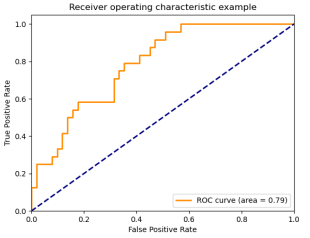


- Step 5: select a detection threshold / decision boundary.
  - Gaussian threshold
  - Robust threshold



- Step 6: **visualize** the detection results as sanity check.
- Step 7: evaluate (with labels or domain knowledge).
  - Threshold dependent: confusion matrix, precision, recall
  - Threshold independent: ROC, PRC

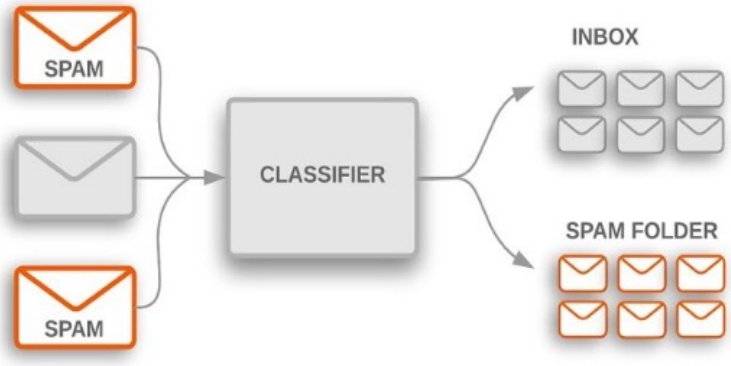




Machine learning: supervised learning

- In ML we train a **model** to learn a **task** on a training **data** set.
- After training, we evaluate the performance on a test set.
- The test set must be drawn from the same data distribution as the training set.

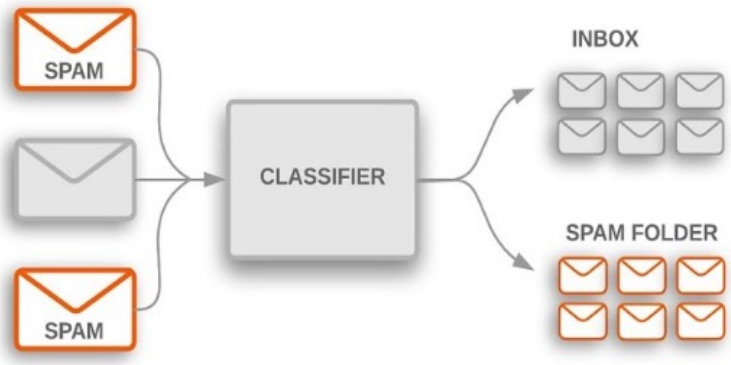
Example: **Model** = random forest
Task = binary classification
Data = emails and spam



Machine learning: supervised learning

- In ML we train a **model** to learn a **task** on a training **data** set.
- After training, we evaluate the performance on a test set.
- The test set must be drawn from the same data distribution as the training set.
- If not: we do not expect a similar performance on the test set.
- Performance is measured using different metrics.
- Which metric? Depends on the data and the task.

Example: **Model** = random forest
Task = binary classification
Data = emails and spam



Training data = both regular emails and spam with LABELS for both

Validation data = labeled regular emails and spam

Test data = regular emails and spam

➔ SUPERVISED AD

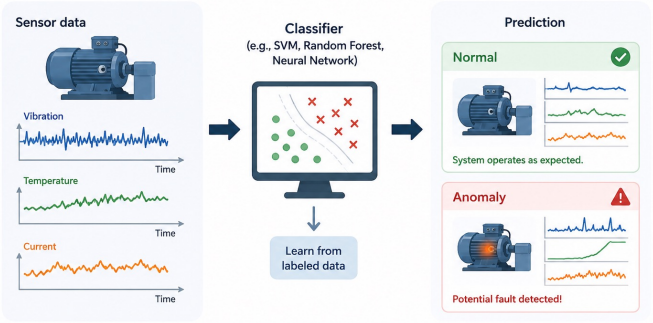
Performance metric e.g confusion matrix.

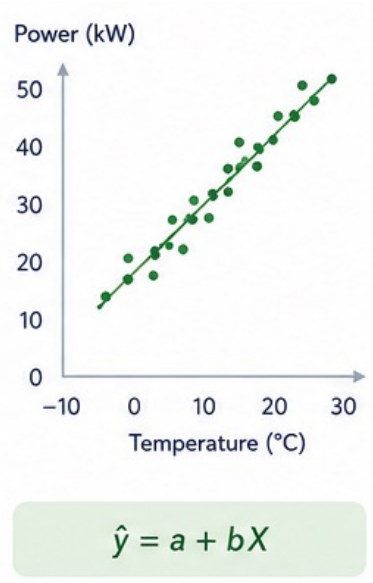
Classification & regression

Fundamental task of supervised learning:

$$x \rightarrow y = f(x)$$

- If y is a categorical variable: Classification.
- If y is a continuous numeric variable: Regression.



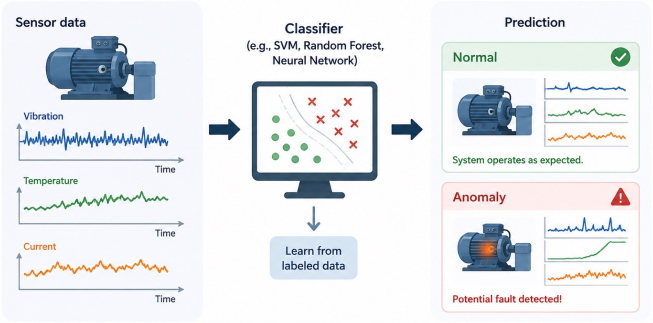


Classification & regression

Fundamental task of supervised learning:

$$x \rightarrow y = f(x)$$

- If y is a categorical variable: Classification.
- If y is a continuous numeric variable: Regression.

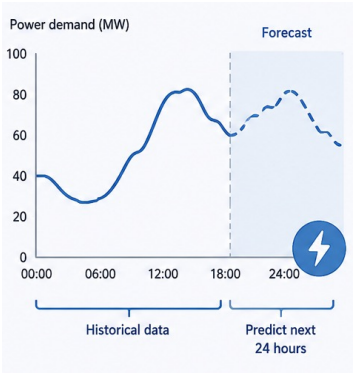


Special cases when y is:

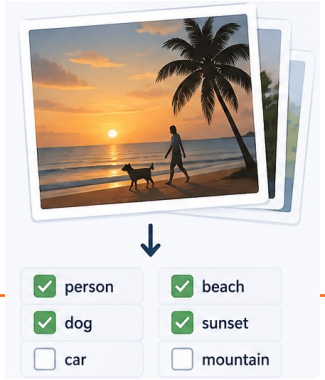
- Ordinal categorical variable or a rank
  - ➢ .customer credit rating → poor/ fair/ good/ excellent



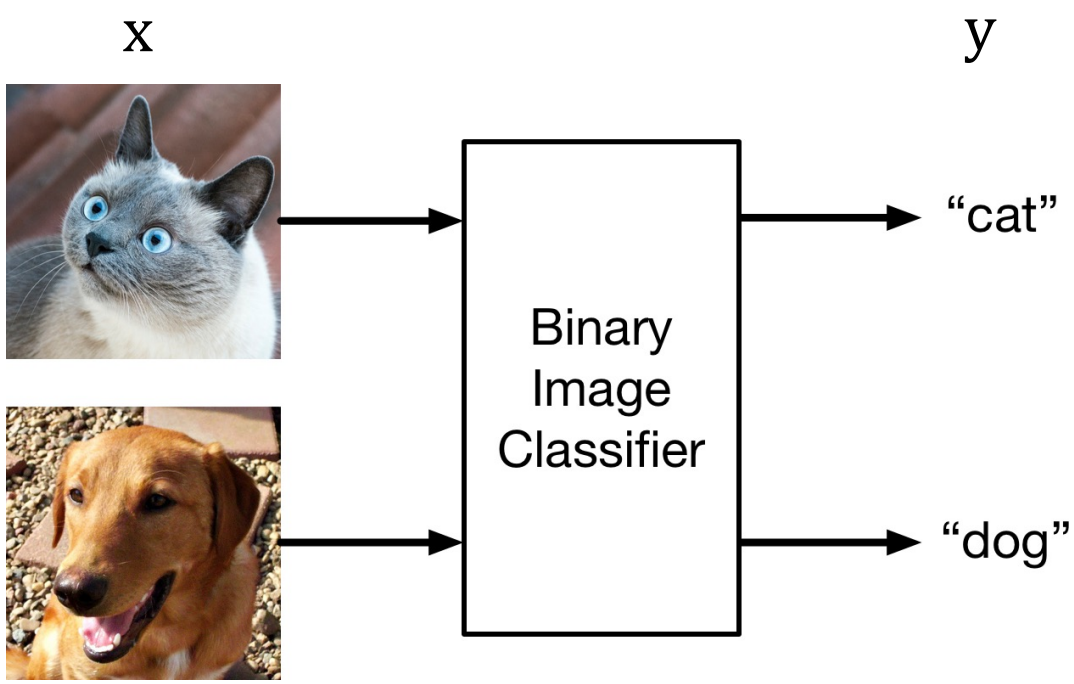
- Sequence of values (e.g. time series)
  - ➢ Electricity demand forecast → Power consumption for the next 24 hours



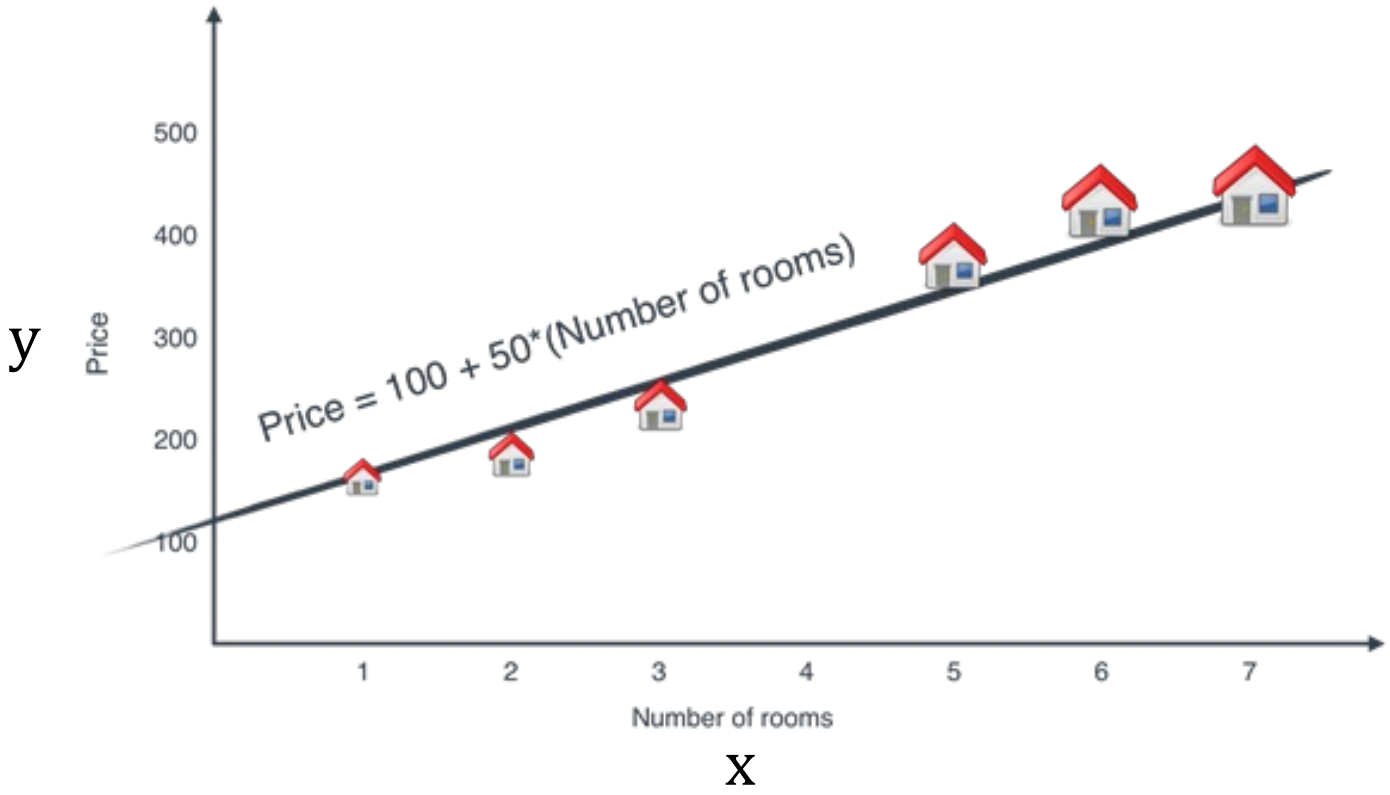
- Vector of arbitrary length (multi-label class)
  - ➢ Photo tagging → {person, dog, beach, sunset}



$$x \rightarrow y = f(x)$$



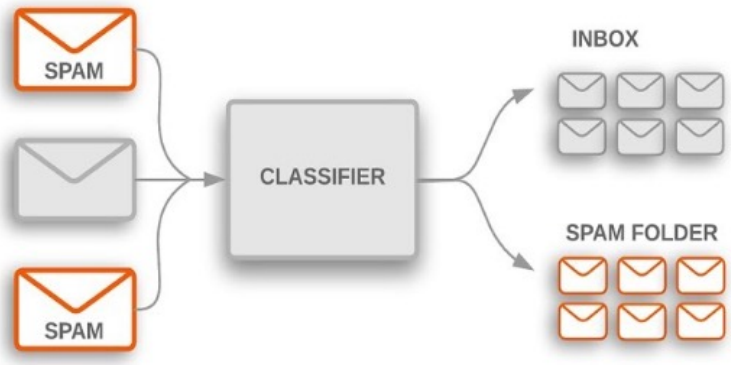
$$x \rightarrow y = f(x)$$



Machine learning: supervised learning

- In ML we train a **model** to learn a **task** on a training **data** set.
- After training, we evaluate the performance on a test set.
- The test set must be drawn from the same data distribution as the training set.
- If not: we do not expect a similar performance on the test set.
- Performance is measured using different metrics.
- Which metric? Depends on the data and the task.

Example: **Model** = random forest
Task = binary classification
Data = emails and spam



Training data = both regular emails and spam with LABELS for both

Validation data = labeled regular emails and spam

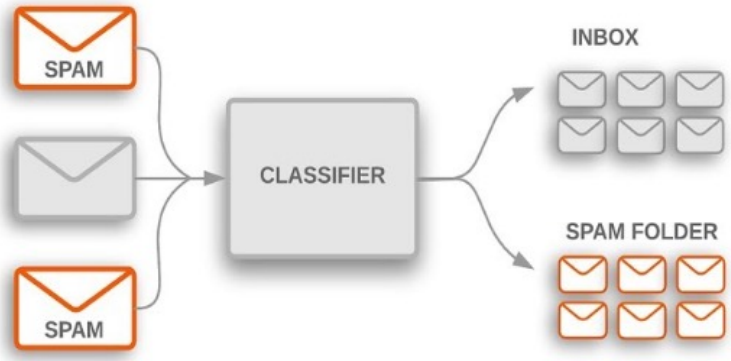
Test data = regular emails and spam

➔ SUPERVISED AD

Performance metric e.g confusion matrix.

Machine learning: unsupervised learning

Example: **Model** = k-means
Task = clustering
Data = emails and spam



Training data = both regular emails and spam without any labels

Validation data = both regular emails and spam

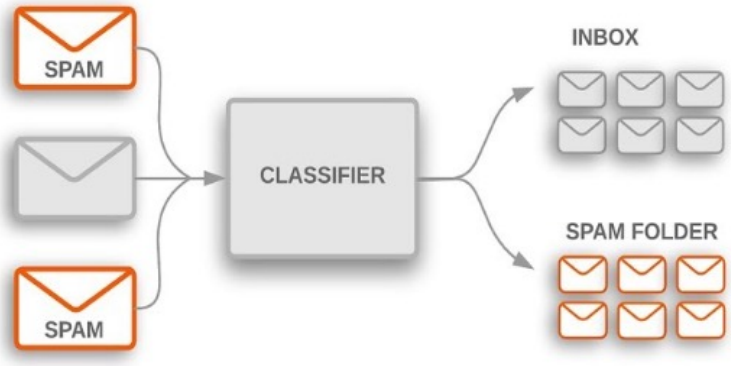
Test data = both regular emails and spam with labels for evaluation

➔ UNSUPERVISED AD

Performance metric e.g confusion matrix.

There is a way in between

Example: **Model** = one-class SVM
Task = novelty detection
Data = emails and spam



Training data = only regular emails, no labels

Test data = both regular emails and spam

➔ NOVELTY DETECTION

Performance metric e.g confusion matrix.

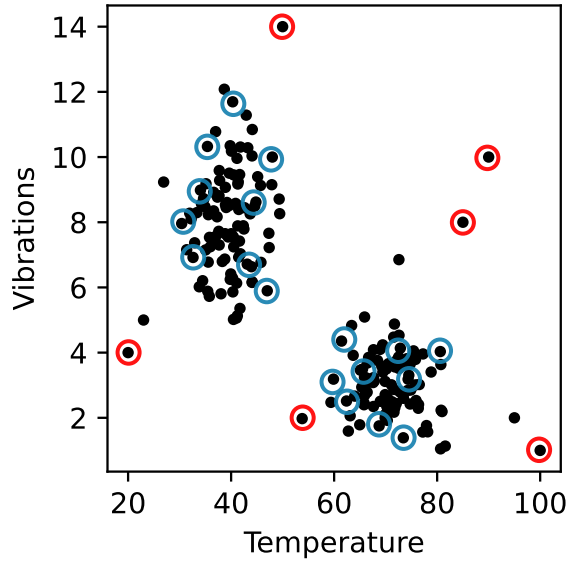
Note: Validation data = we may need both types!

Different approaches to anomaly detection

Supervised methods: learn to represent normal and abnormal data based on previous examples.

➔ need labeled examples of normal and abnormal.

- 👍 high accuracy, easier to validate, anomaly class is clear.
- 👎 Disadvantage: high imbalance, high requirements from the data novelties cannot be discovered.



Different approaches to anomaly detection

Supervised methods: learn to represent normal and abnormal data based on previous examples.

➔ need labeled examples of normal and abnormal.

👍 high accuracy, easier to validate, anomaly class is clear.

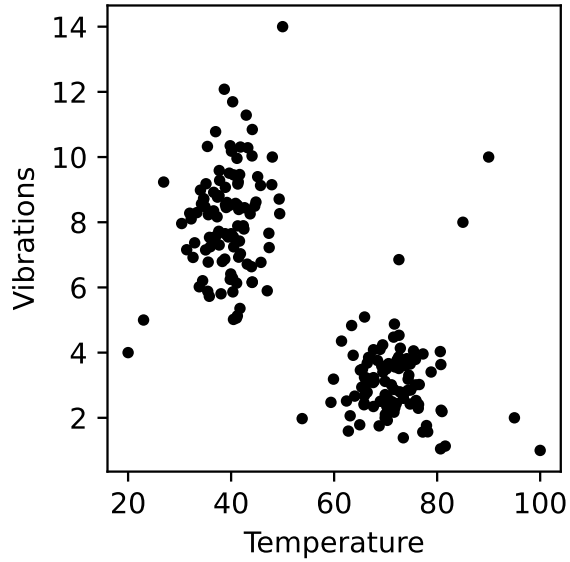
👎 Disadvantage: high imbalance, high requirements from the data, novelties cannot be discovered.

Unsupervised methods: detect patterns in the data that correspond to normal/ abnormal behaviour without any labels.

➔ no labels needed.

👍 Low requirements on data.

👎 Harder to achieve high performance, hard to evaluate.



Different approaches to anomaly detection

Supervised methods: learn to represent normal and abnormal data based on previous examples.

➔ need labeled examples of normal and abnormal.

👍 high accuracy, easier to validate, anomaly class is clear.

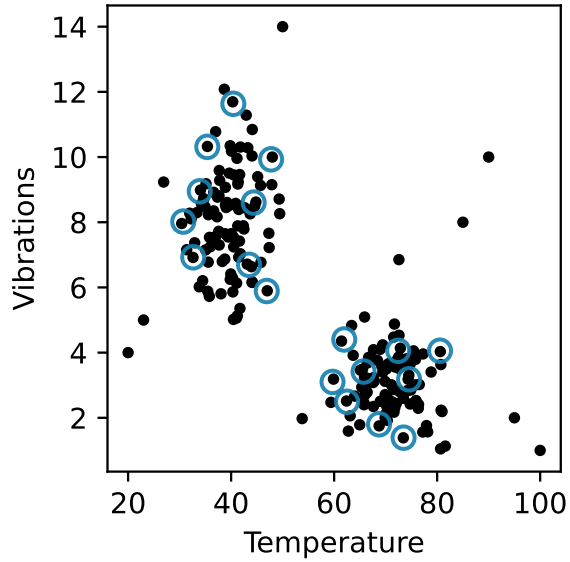
👎 Disadvantage: high imbalance, high requirements from the data, novelties cannot be discovered.

Unsupervised methods: detect patterns in the data that correspond to normal/ abnormal behaviour without any labels.

➔ no labels needed.

👍 Low requirements on data.

👎 Harder to achieve high performance, hard to evaluate.



Novelty detection (most common): learn to represent normal data and compare unknown patterns to the representation.

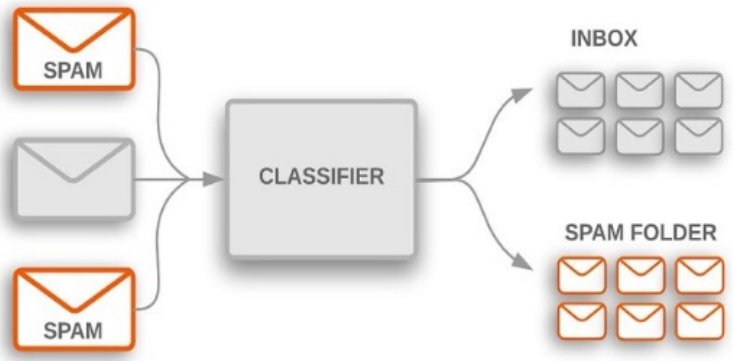
➔ need labels only for normal examples.

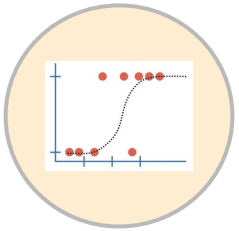
👍 normal data is usually abundant, detection independent of anomaly type.

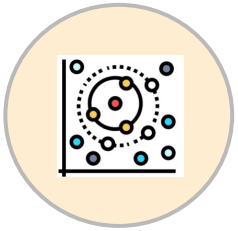
👎 Hard to validate, anomaly class unclear.

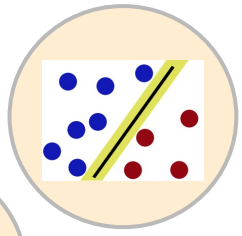
Supervised AD methods

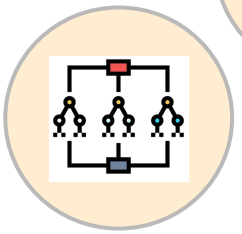
- Labeled normal and abnormal samples.
- Binary classification methods. Examples:
  - Logistic regression (Example)
  - K-nearest neighbors
  - Support vector machines (Example)
  - Classification trees
  - XGBoost
  - Classification NNS
  - …











Often used as «best case baseline»

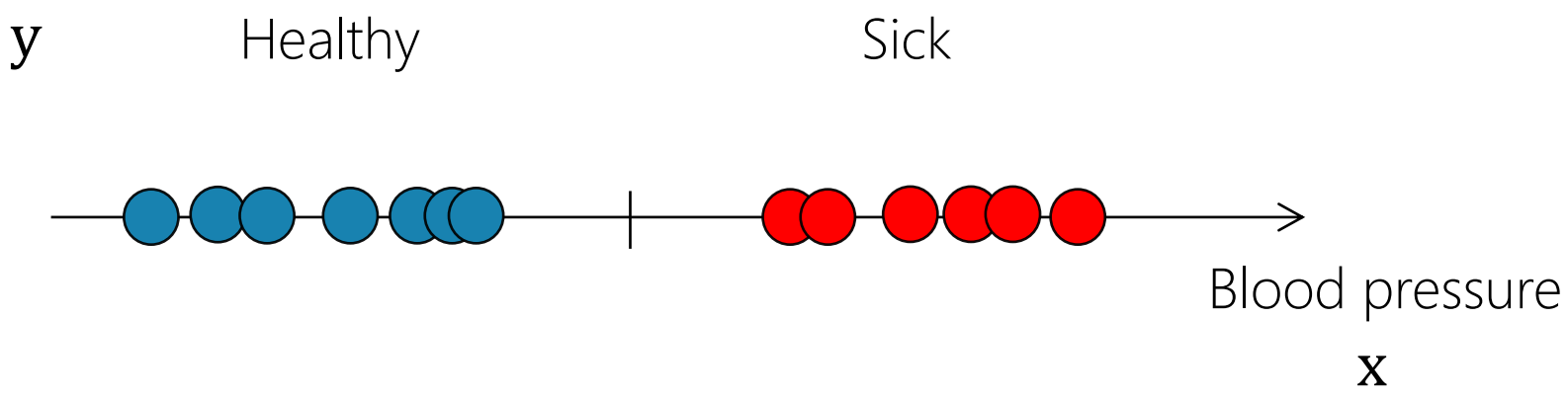
Image sources: https://www.linkedin.com/pulse/binary-classification-pavan-kumar-4iqbc/ https://www.vecteezy.com/; https://www.reddit.com/r/MachineLearning

Supervised AD: logistic regression

- Learn from labeled examples, if there are enough.
- Classify new points.

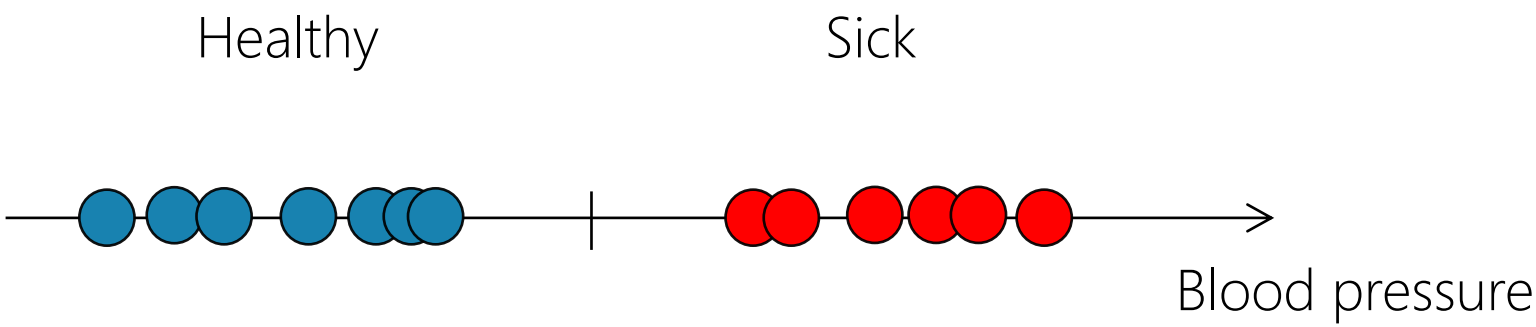
➔ Where is the correct decision boundary?

$$x \rightarrow y = f(x)$$



Question

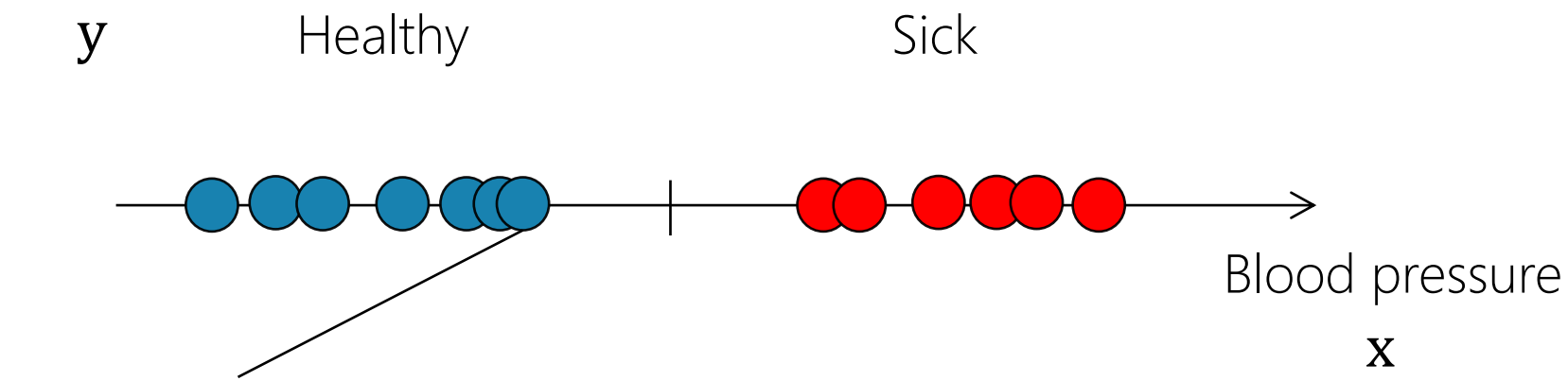
- How would you formulate the same problem
  - as an unsupervised task?
  - as a novelty detection task?



- Learn from labeled examples, if there are enough.
- Classify new points.

➔ Where is the correct decision boundary?

$$x \rightarrow y = f(x)$$



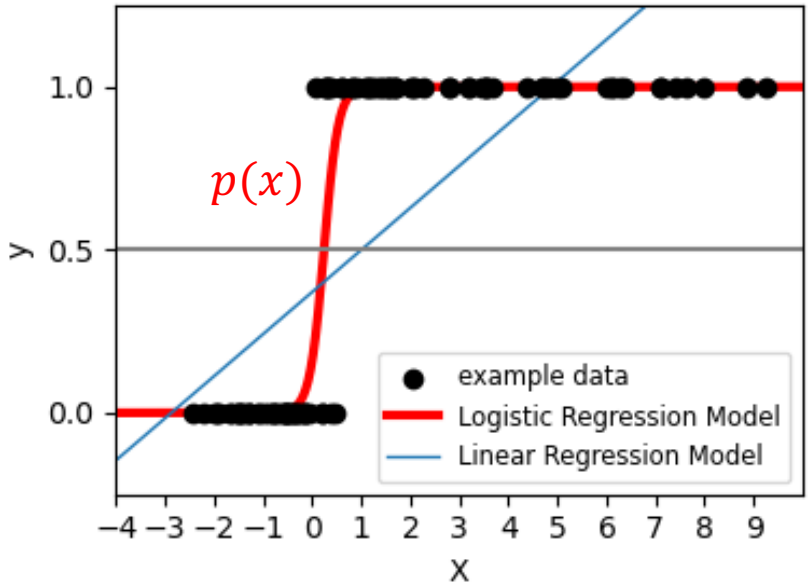
Each point k consists of

- a set of m input variables $x_{1,k}$ … $x_{m,k}$ → blood pressure
- and a binary outcome variable $\mathbf{y_k}$ → healthy / sick

Logistic Regression

- Each point k consists of
  - a set of m input variables $x_{1,k}$ ... $x_{m,k}$
  - and a binary outcome variable $\mathbf{y_k}$

- use the dataset to create a predictive model of the outcome variable
- Define $\mathbf{p}$ the probability that $\mathbf{y_k} = 1$.
- Define $\mathbf{t}$ the log odds:



$$\mathbf{t} = \log \frac{p}{1-p} = \underbrace{\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_m x_m}_{\boldsymbol{\beta}\cdot\mathbf{x}}$$

Solving for the probability p :

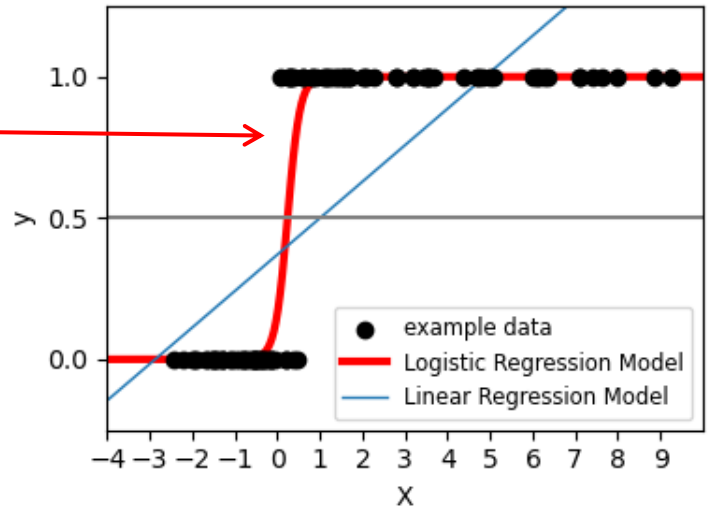
$$p(\mathbf{x}) = \frac{e^{\boldsymbol{\beta}\cdot\mathbf{x}}}{1+e^{\boldsymbol{\beta}\cdot\mathbf{x}}} \quad \begin{cases} 0 & for \quad x \to -\infty \\ 1 & for \quad x \to \infty \end{cases}$$

β determines how fast

Image source: https://scikit-learn.org

Find the parameters $\boldsymbol{\beta}$ that create a $p(\mathbf{x})$ that is most likely to fit the data:

$$p(x) = \frac{e^{\boldsymbol{\beta}\cdot x}}{1+e^{\boldsymbol{\beta}\cdot x}}$$



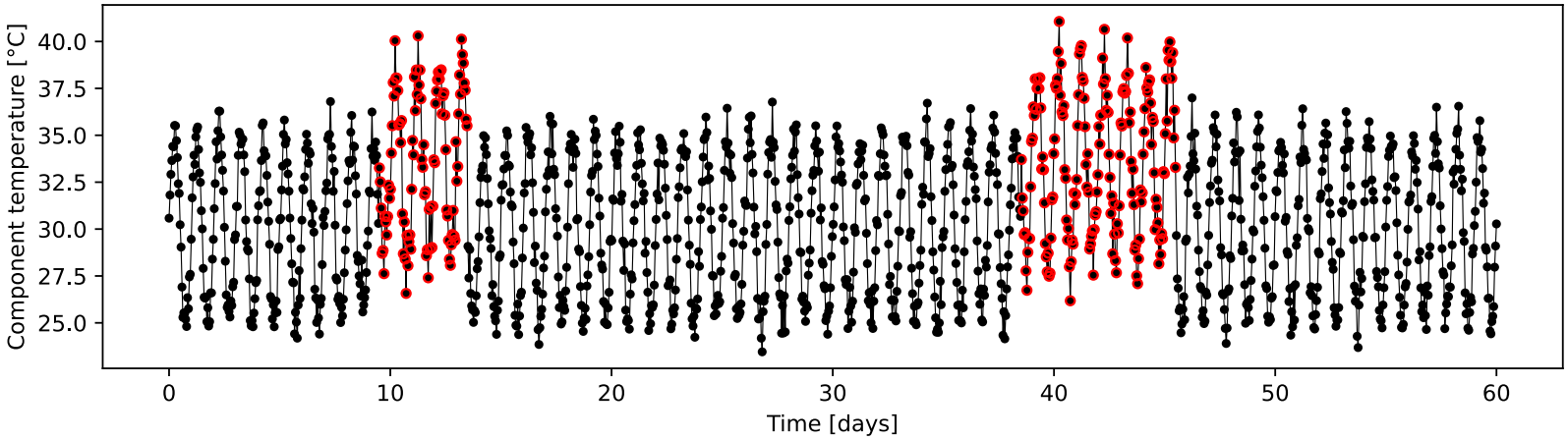
The optimal β are found by maximizing the log-likelihood:

$$l = \sum_{k=1}^{N} y_k \log[p(x_k)] + \sum_{k=1}^{N} (1 - y_k) \log[\,1 - p(x_k)]$$

$$\frac{\partial l}{\partial \beta_m} = 0 \quad \rightarrow \quad \boldsymbol{\beta}^*$$

Example

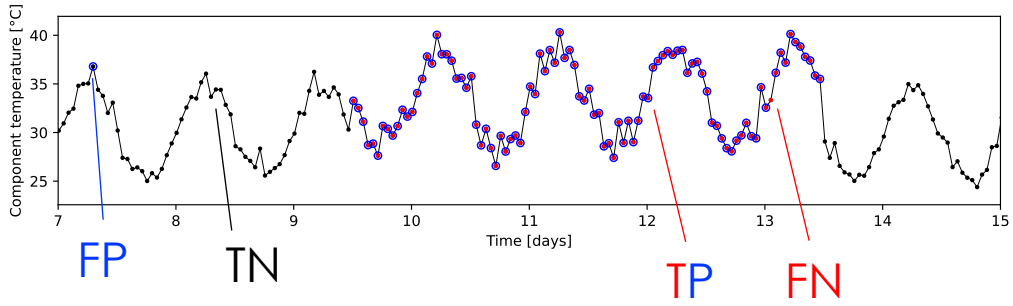
- Component temperature vs. time (hourly sensor values).
- The component is affected by the ambient temperature.
- Certain periods suffer faults that are then repaired.



Evaluation

- Binary classifier predictions:
  - Positive: classifier predicts “anomaly”
  - Negative: classifier predict “normal”

  (Think of Covid…)



- Compare to true label:
  - TP **True** positive: classifier detects **correctly** an anomaly.
  - TN **True** Negative: classifier detects **correctly** a normal sample.
  - FP **False** positive: classifier **mis**classifies a true normal sample.
  - FN **False** negative: classifier **mis**calssifies a true anomaly.

➔ Confusion matrix

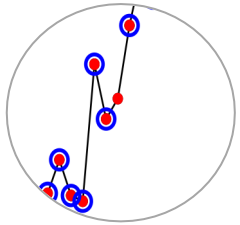
		PREDICTED: Positive	PREDICTED: Negative
ACTUAL	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

- Tradeoff between 2 types of misclassifications.
  - Low detection threshold ➔ many false alarms (“false positives”)
  - High detection threshold➔ many missed detections (“false negatives”)

What is relevant to ask:

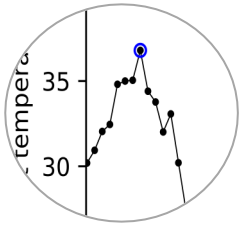
- If there is an anomaly, how probable I am to detect it?

$$Recall = TPR = sensitivity = \frac{TP}{TP + FN}$$



- If there is no anomaly, what is the probability that I classify it wrongly?

$$FPR = 1 - specificity = \frac{FP}{TN + FP}$$



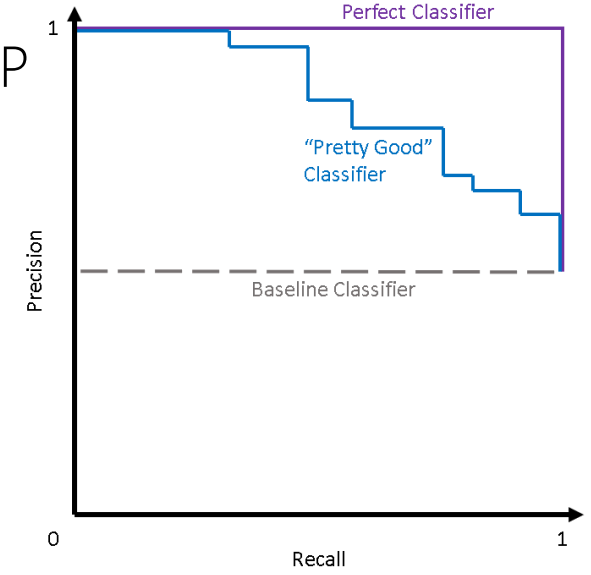
- If I detected an anomaly, what is the probability that it is indeed an anomaly?

$$Precision = \frac{TP}{TP + FP}$$

Threshold independent evaluation metrics

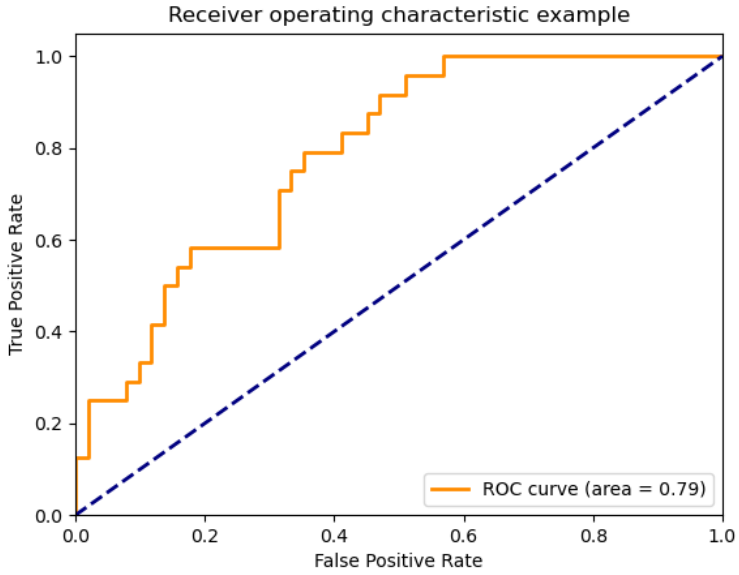
PRC plot➔ AUPRC=AP

$$Precision = \frac{TP}{TP + FP}$$



$$Recall = TPR = \frac{TP}{TP + FN}$$

ROC plot➔ AUROC



$$FPR = \frac{FP}{TN + FP}$$

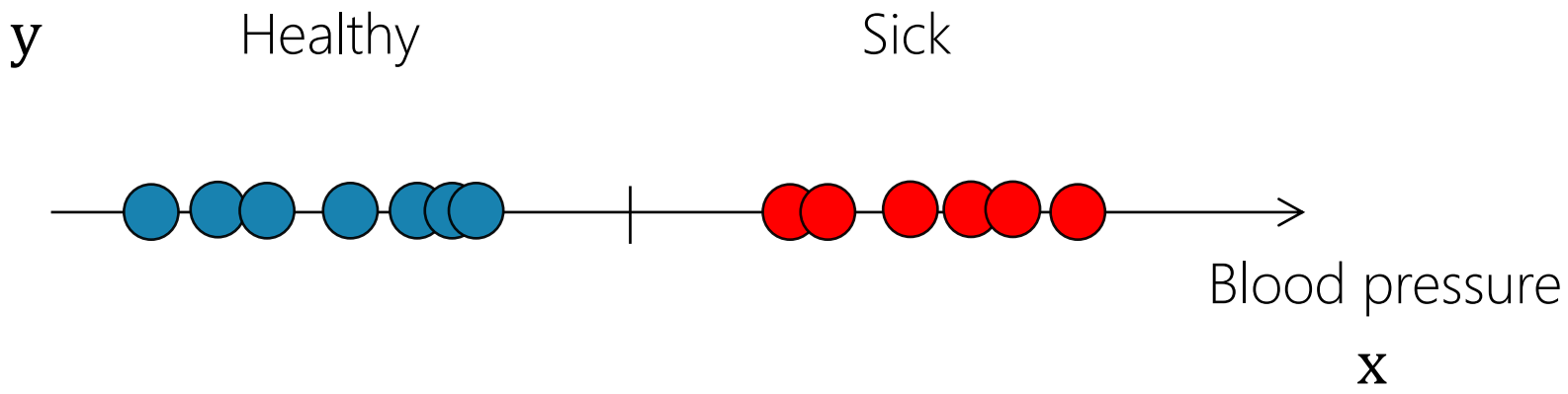
WPI_AnoDet_02a_AD_supervised_log_reg_classifier

- Learn from labeled examples.
- Classify new points.

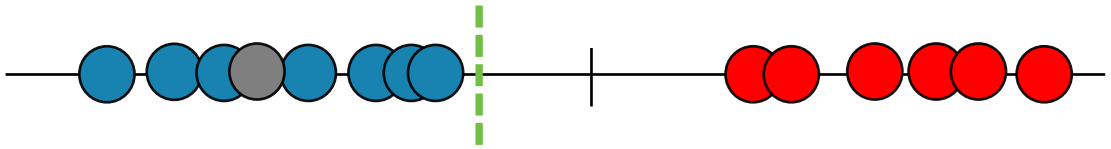
➔ Where is the correct decision boundary?

SVM

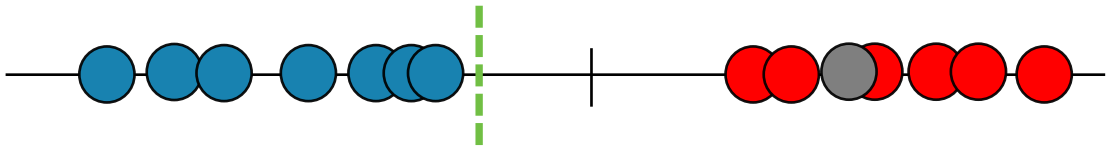
$$x \rightarrow y = f(x)$$



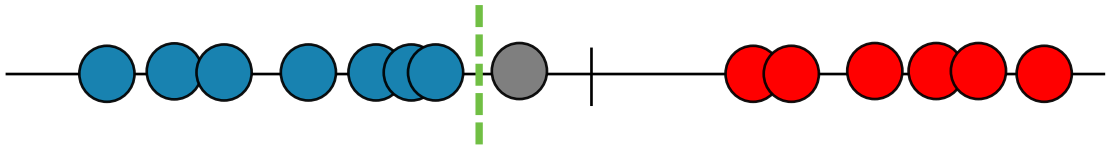
How to select a threshold?



clear

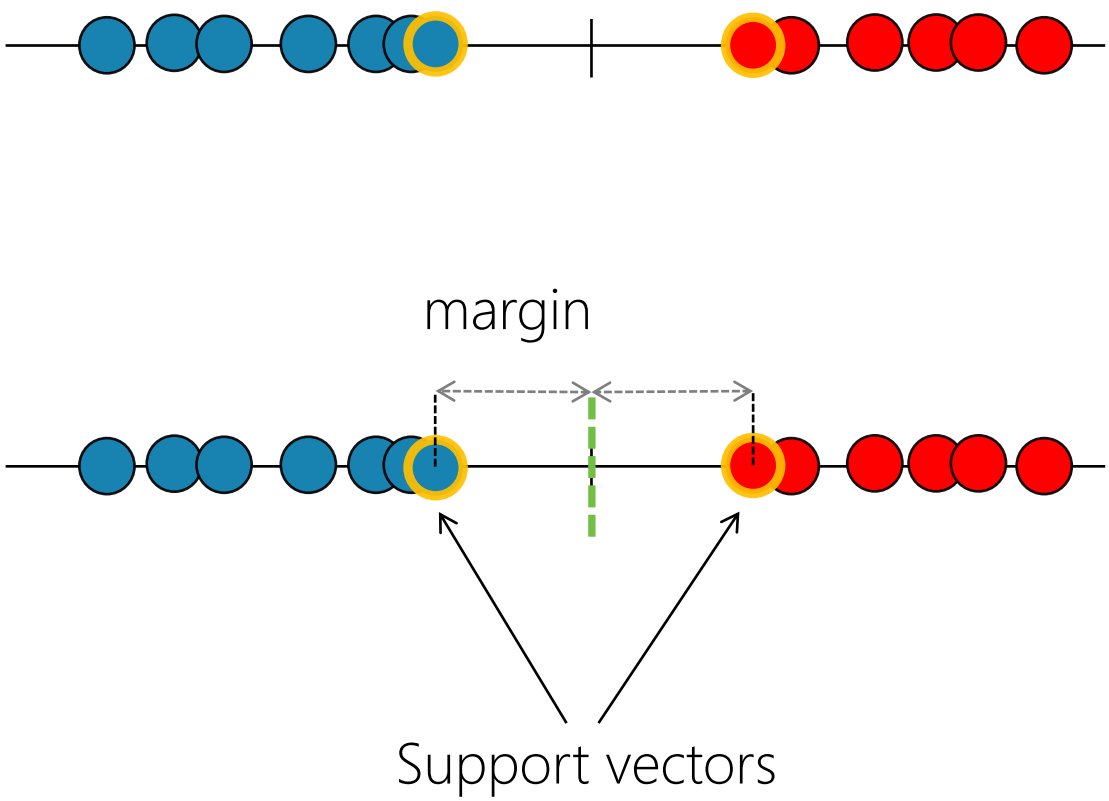


clear



Unclear…

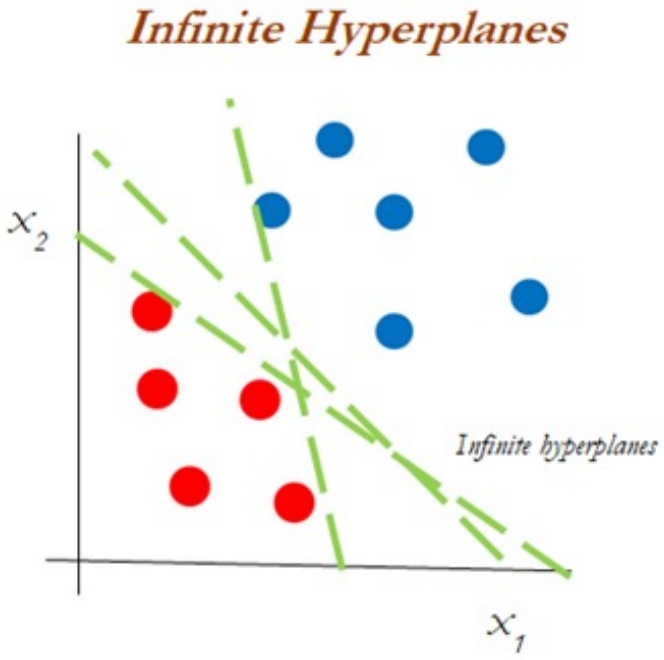
To decide on the correct threshold look at observations at the edge:

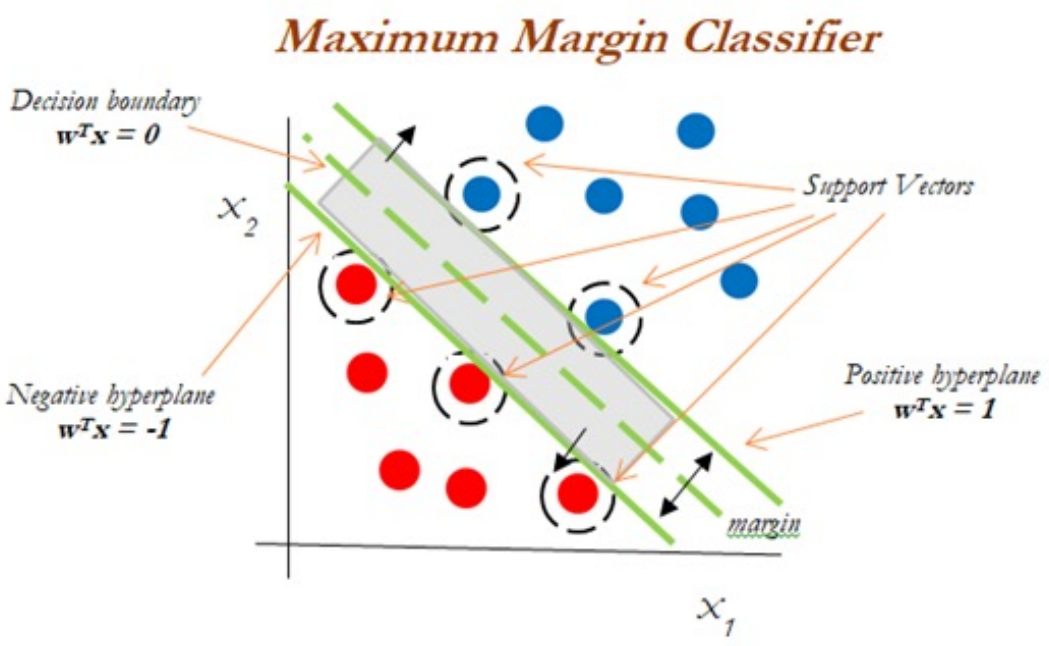


Choose the maximal margin

Higher dimensional examples

Example: x_1 = blood pressure, x_2 = weight. Predict: healthy/sick.



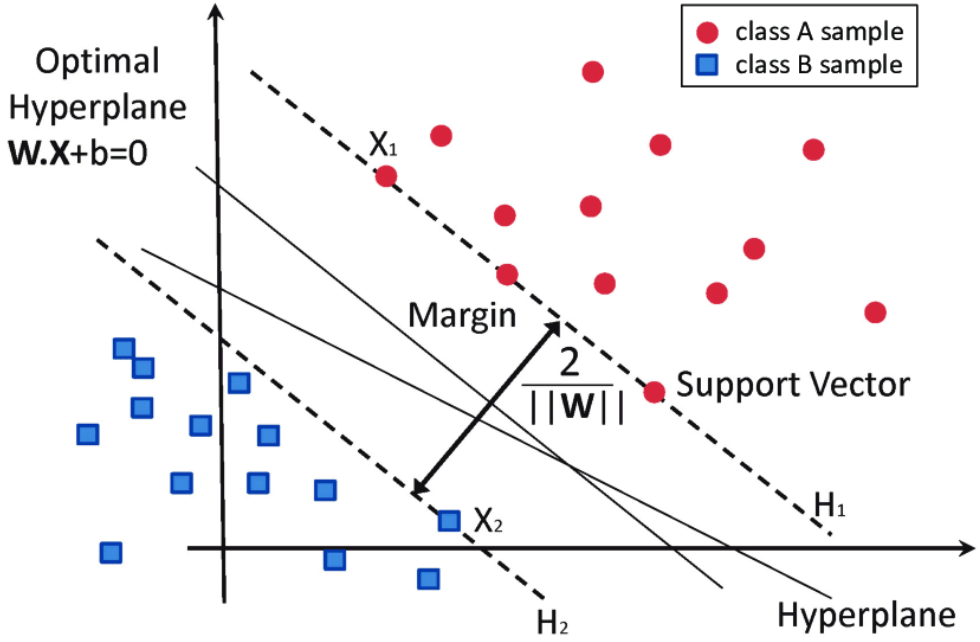


In n dimensional feature space:

- A sample is a point in n-dimensional space
- The separating hyperplane is n-1 dimensional
- The support vectors are n-1 dimensional

Image source: https://www.packtpub.com/en-us/

Find the maximum margin classifier



A classifier is defined by

$$y = 1 \quad if\ x\ in\ class\ A\ (sick)$$
$$y = -1\ if\ x\ in\ class\ B\ (healthy)$$

SVM finds the parameters $\vec{w} = (w_1, w_2)$ and b by solving the optimization problem

$$\min_{w,b} \frac{1}{2}\|\vec{w}\|^2 \qquad \text{such that} \qquad y_i(\vec{w} \cdot \vec{x}_i + b) \geq 1 \quad \forall \vec{x}_i$$

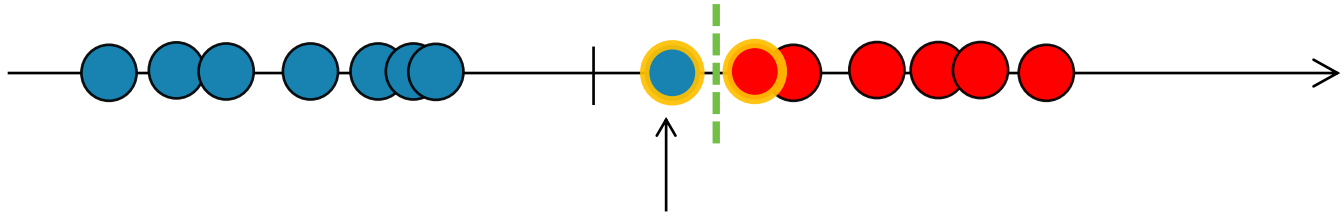
Labels (1 or -1)

The normal to the hyperplane

This maximizes the margin $\frac{2}{\|\vec{w}\|}$ while learning a suitable separating hyperplane $\vec{w}$.

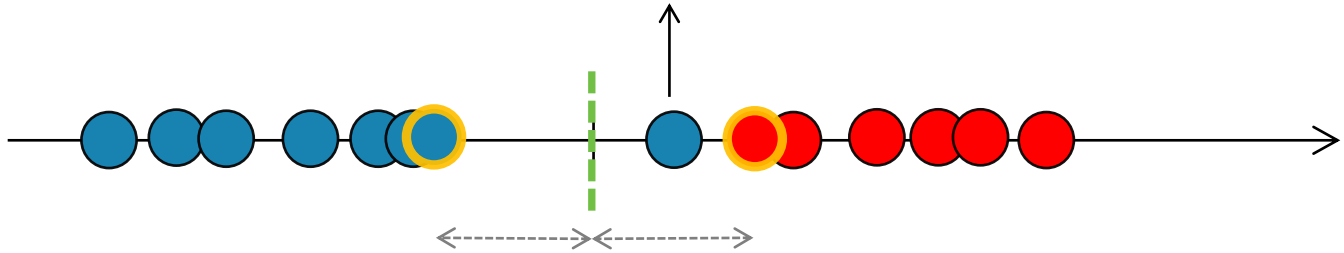
Soft margin Support Vector Classifier

Problem: maximal margin is sensitive to outliers



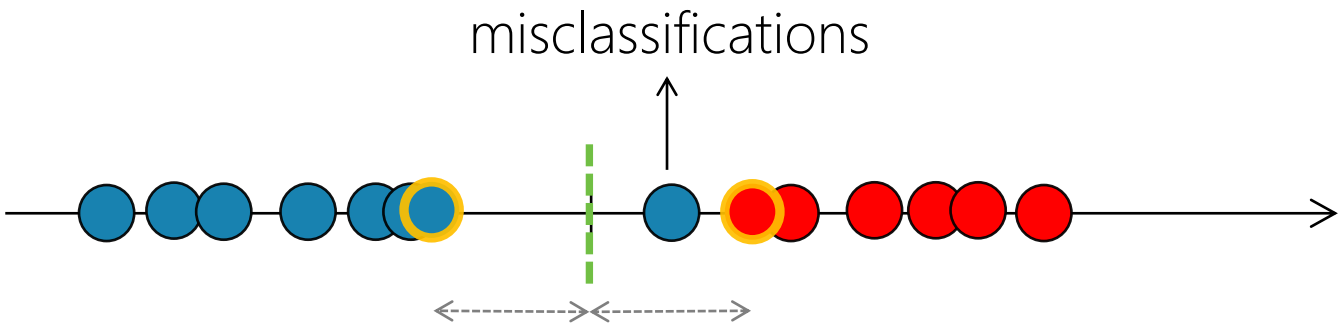
outlying sample:
far but still healthy

Solution: allow for some misclassifications



Select the support vectors
to allow for soft margins

Soft margin Support Vector Classifier



Select the support vectors
to allow for soft margins

↓

Hyperparameter: no. of allowed miscalssifications

↓

Determine through cross validation

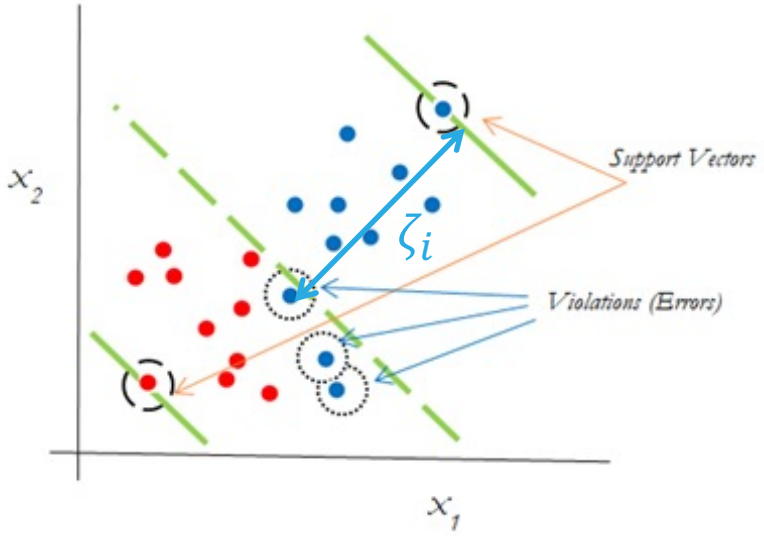
Soft margin Support Vector Classifier

$$\min\left\{\frac{1}{2}\|\vec{w}\|^2 + C\sum_{i=1}^{n}\zeta_i\right\} \quad \text{such that} \quad y_i(\vec{w}\cdot\vec{x}_i + b) \geq 1 - \zeta_i \;\; \forall\vec{x}_i$$

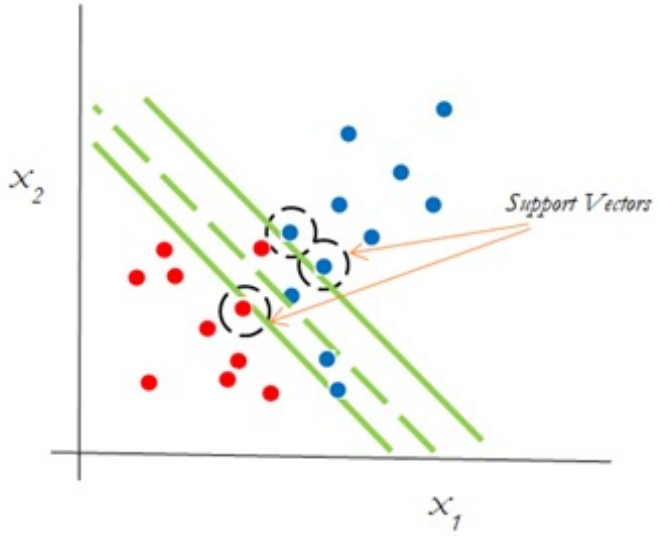
Regularization param.
Panelizes misclassifications

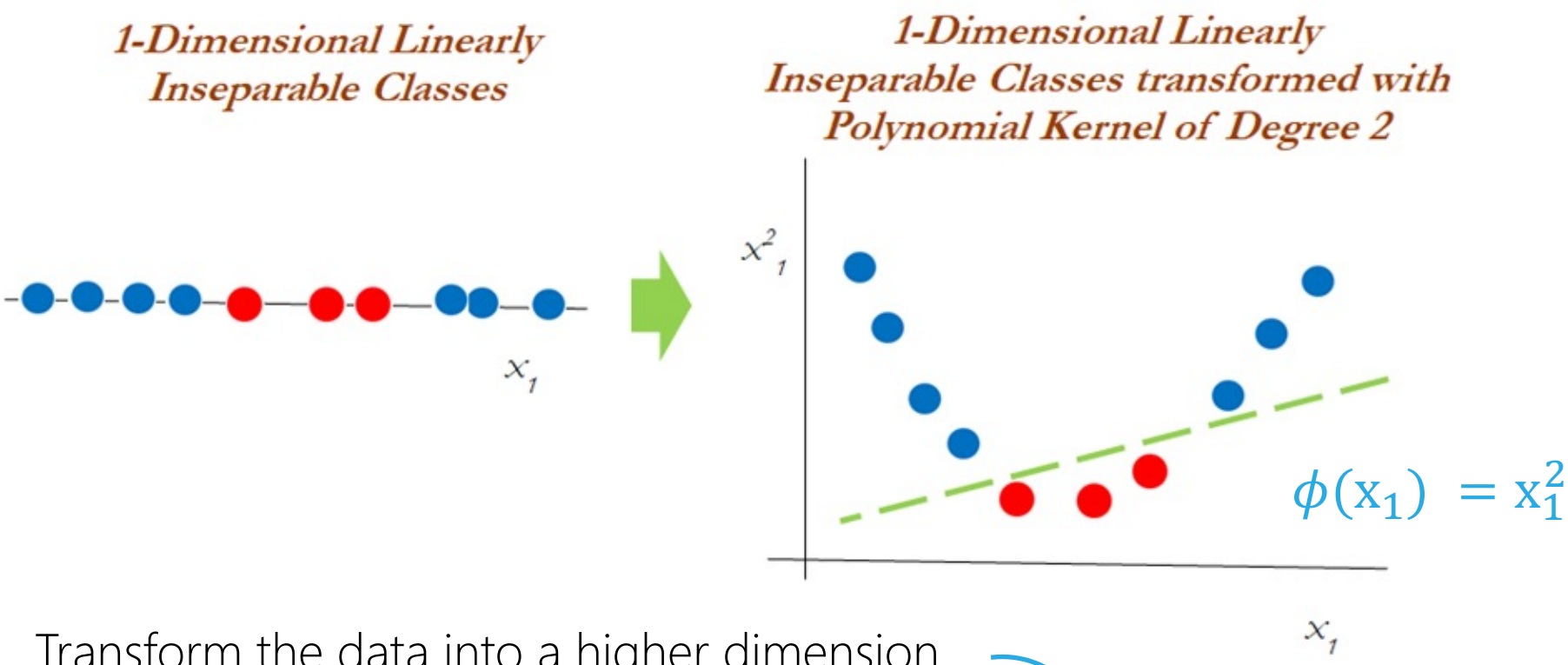
$\neq 0$ only for misclassified samples or within margin

Small C: allow many errors



Large C: avoid many errors





1. Transform the data into a higher dimension.
2. Find a separating hyperplane in the new space.

$$\min\left\{\frac{1}{2}\|\vec{w}\|^2 + C\sum_{i=1}^{n}\zeta_i\right\} \quad \text{such that} \quad y_i(\vec{w}\cdot\phi(\vec{x}_i) + b) \geq 1 - \zeta_i \;\; \forall\vec{x}_i$$

Image source: https://www.packtpub.com/en-us/

1. Transform the data into a higher dimension.
2. Find a separating hyperplane in the new feature space.

Effectively, express the transformation as a kernel.

Linear kernel:

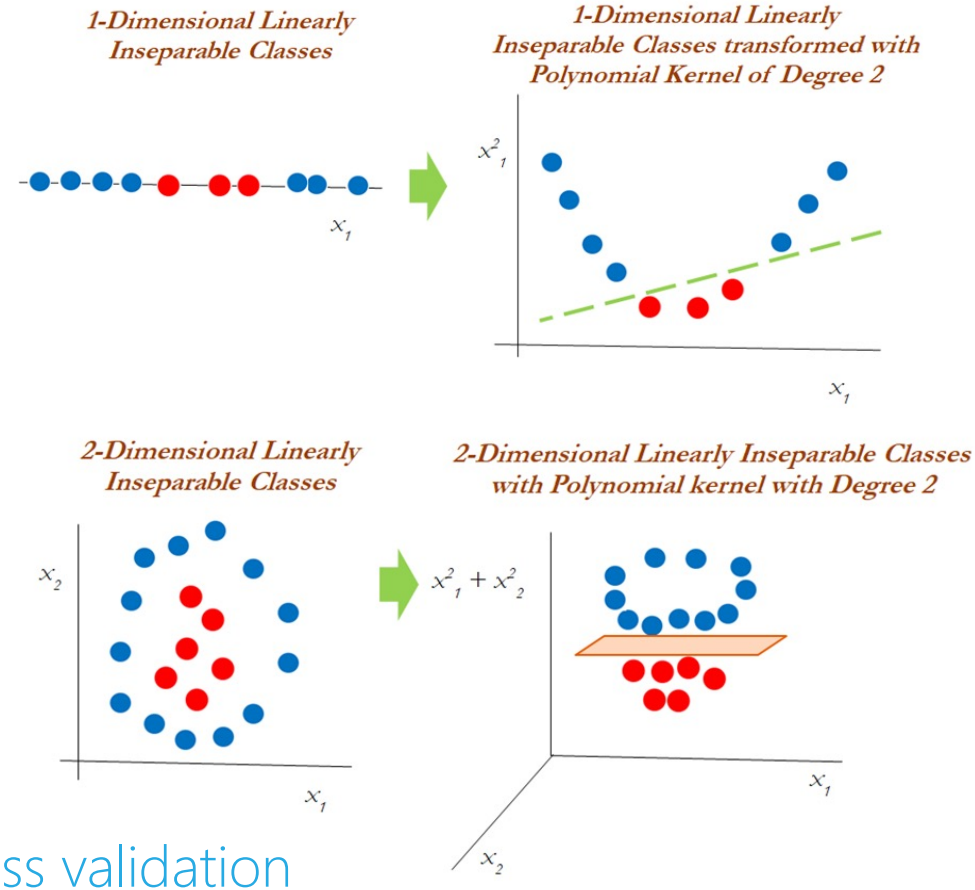
$$K(\vec{x}_i, \vec{x}_j) = \vec{x}_i \cdot \vec{x}_j$$

Polynomial kernel:

$$K(\vec{x}_i, \vec{x}_j) = (r + \gamma \vec{x}_i \cdot \vec{x}_j)^d$$

Radial basis function (Gaussian) kernel:

$$K(\vec{x}_i, \vec{x}_j) = e^{-\gamma \|\vec{x}_i - \vec{x}_j\|^2}$$



Hyperparameters ➔ determine with cross validation

Image source: https://www.packtpub.com/en-us/

Exercise: supervised AD with SVM

***class* sklearn.svm.SVC(*, *C*=*1.0*, *kernel*='*rbf*', *degree*=*3*, *gamma*='*scale*')**

Linear → straight separation line

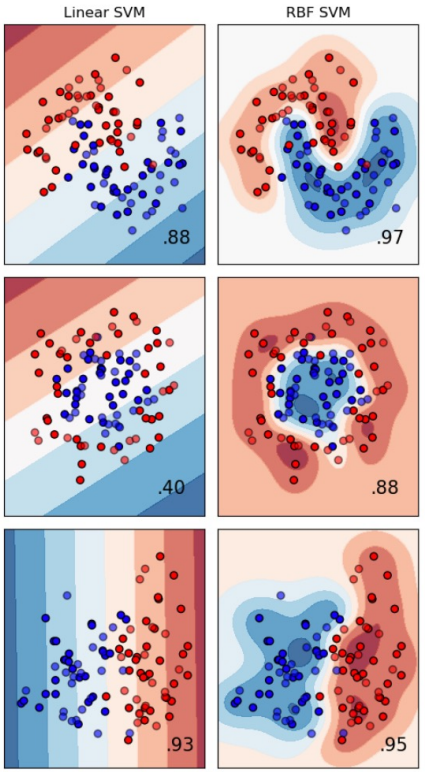
SVC(kernel="linear", C=0.025)

Polynomial → smooth separation curve of limited complexity

SVC(kernel="poly", degree =4, coef0=1, gamma=2, C=1)

RBF → highly flexible boundary that can bend locally around the data

SVC(kernel="rbf", gamma=2, C=1)



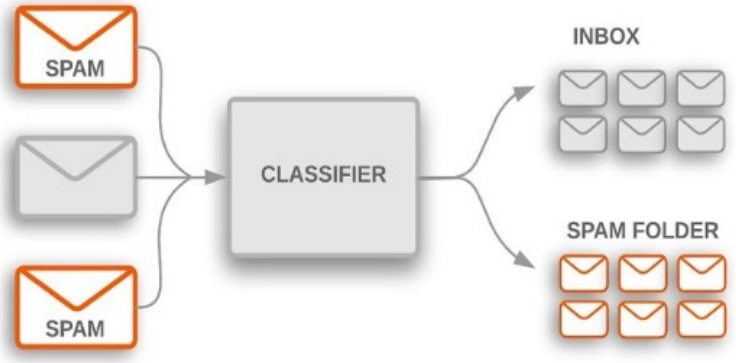
WPI_AnoDet_02b_exercise_AD_supervised_svm_classifier

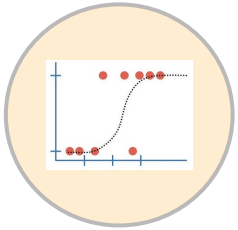
Solution: supervised AD with SVM

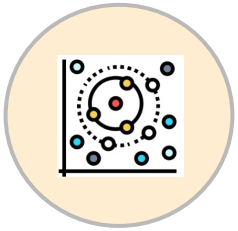
- WPI_AnoDet_02c_solution_AD_supervised_svm_classifier

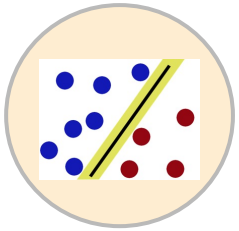
Recap Supervised AD methods

- Labeled normal and abnormal samples.
- Binary classification methods. Examples:
  - ✓Logistic regression
  - K-nearest neighbors
  - ✓Support vector machines
  - Classification trees
  - XGBoost
  - …









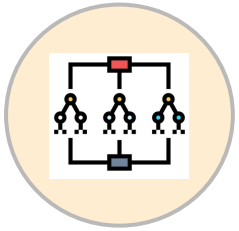
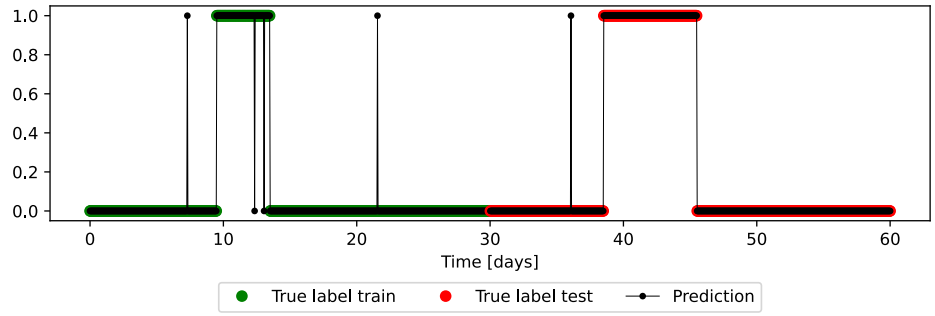


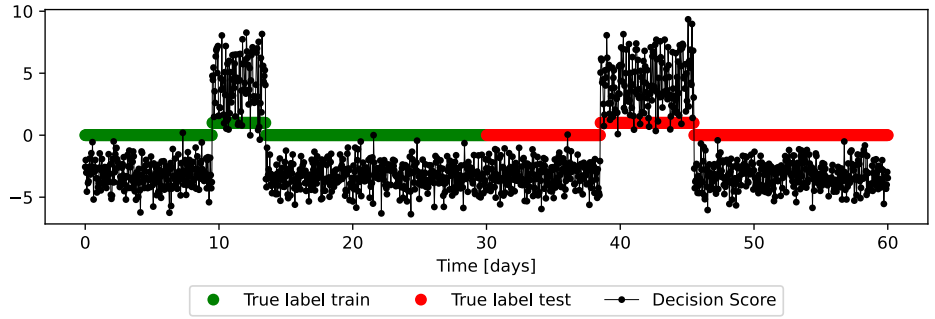
Image sources: https://www.linkedin.com/pulse/binary-classification-pavan-kumar-4iqbc/ https://www.vecteezy.com/; https://www.reddit.com/r/MachineLearning

Understand the algorithm

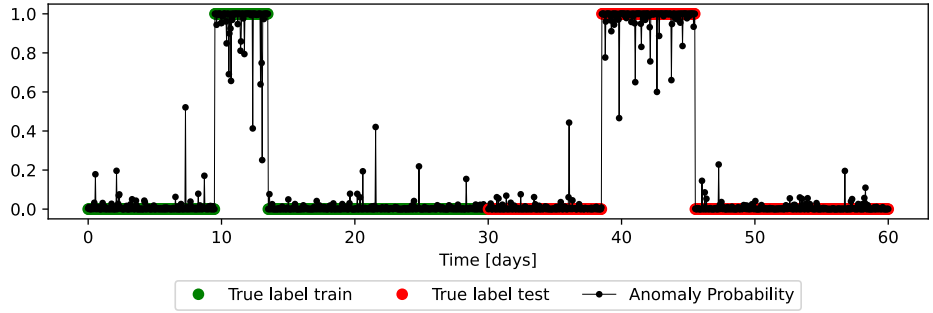
- Final classification output



- Decision score

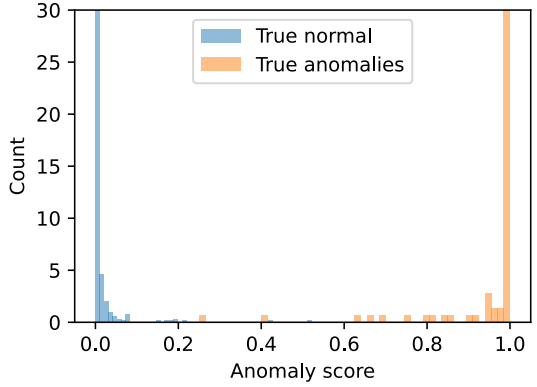


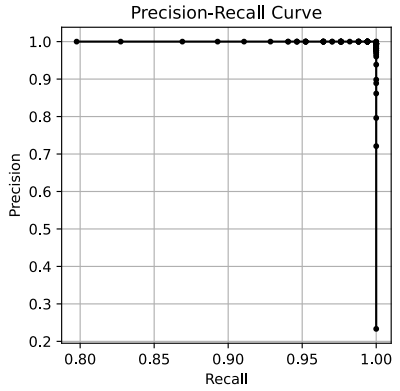
- Anomaly probability



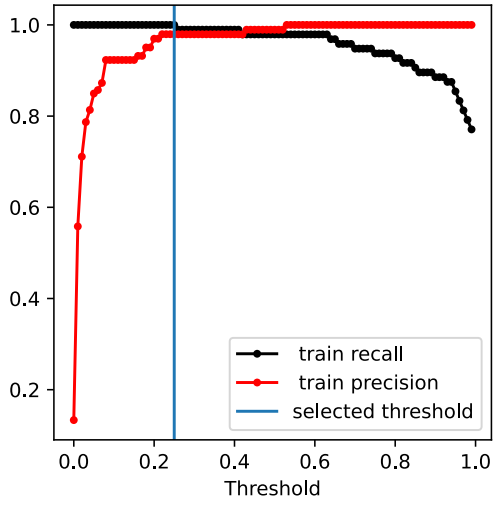
Supervised AD: use labeled data to select a threshold

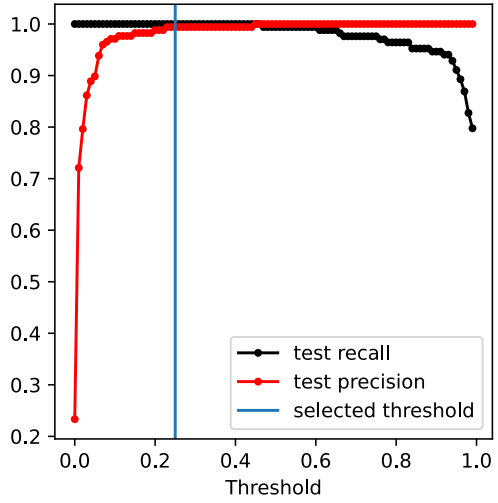
- Anomaly scores distribution





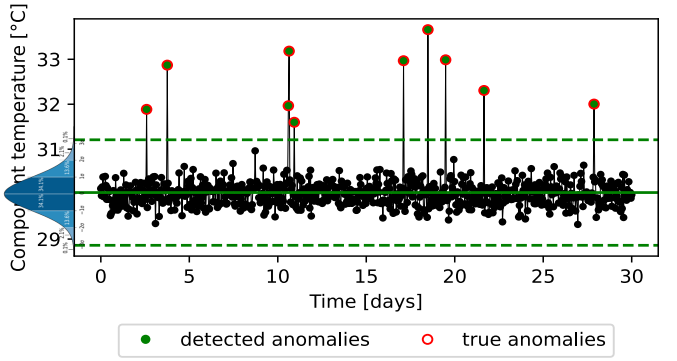
- Select threshold on training data based on use-case





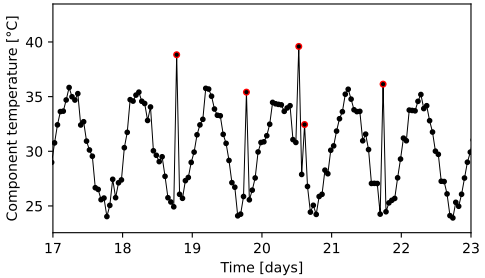
Recap

- AD in a univariate variable ➔ statistical threshold



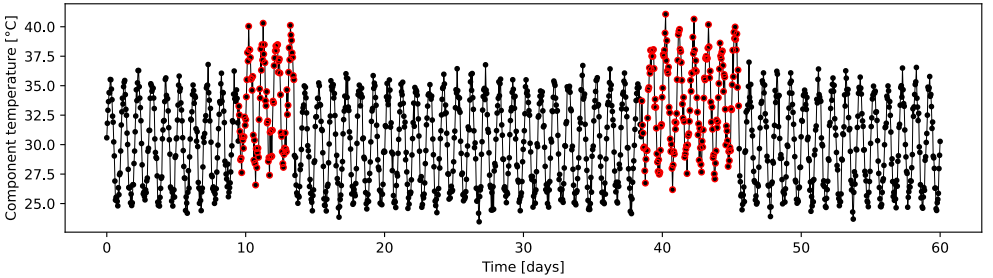
- AD in multivariate data ➔ (linear) regression, residuals as scores.





- Supervised AD with logistic regression.

- Supervised AD with SVM.



- Evaluation with precision-recall and ROC

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

Assignments

- Read about all data sets
- Organize into groups
- Discuss potential assignments
- Note:
  - Some data sets look “simple” but involve more task **designing**
  - The task **difficulty** will be an evaluation criterium
  - Applying techniques and **learnings** from class is important